Operator-Consistency Failure and its Resolution in Primitive-Node Pseudo-Spectral Time Stepping of Variable-Coefficient Anelastic Flow

Kiriko[a,*]

[a] *Department of Astronomy, Tsinghua University, China*

ARTICLE INFO

Keywords:
pseudo-spectral methods
anelastic approximation
stellar pulsation
operator consistency
g-modes

ABSTRACT

Nodal pseudo-spectral discretisations of variable-coefficient anelastic flow that apply the elliptic operator *matrix-free* (as a composition of differentiation, pointwise multiplication, and pointwise division kernels, without constructing the assembled matrix) exhibit a structural operator-consistency gap at the discrete level. We identify, quantify, and resolve this gap, and show that it reflects a more general property: matrix-free nodal pseudo-spectral discretisations of variable-coefficient elliptic generalised eigenproblems converge, in the continuous limit, to a local surrogate operator rather than the intended global elliptic inverse. When the scheme is reduced to an explicit second-order oscillator by eliminating the momentum/pressure coupling, initialising it with an exact g-mode eigenvector of its own discrete eigenproblem produces a per-step deviation of $\mathcal{O}(10^{-4})$ that is independent of grid resolution, independent of time-step refinement, and independent of the boundary-condition formulation; under further factorisation of the elliptic operator the reduced scheme acquires negative eigenvalues and diverges exponentially within a single oscillation period. Production pseudo-spectral anelastic codes avoid the latter regime by retaining pressure and solving a global variable-coefficient Poisson equation at each substage; the present work shows that this pressure solve acts as an *implicit* assembly of the same elliptic inverse. The failure arises from replacing the inverse of a global elliptic operator by a pointwise scaling: a structural inconsistency, not a singularity or a truncation effect, that persists under both grid refinement and small-perturbation limits of the physics amplitude. The associated relative consistency gap is shown to admit a resolution-independent lower bound that scales linearly in $\|\rho_0'\|_\infty$ in the small-perturbation limit and is structural rather than truncation-driven. The failure is resolved by assembling $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$ once at setup and applying it as a per-wavenumber dense matrix-vector product at every RK4 substage; this preserves any eigenvector of the assembled eigenproblem to within $\mathcal{O}(\epsilon_{\mathrm{mach}}\,\kappa(Q))$ round-off per step. The CPU reference implementation achieves a per-step deviation of 5×10^{-18}, while a GPU-accelerated port maintains accuracy at the 3×10^{-15} level; the spatial machinery is externally validated against the GYRE stellar-pulsation code to 9.1×10^{-9} relative error on a Lane–Emden $n = 3$ polytrope at $N_r = 96$. At $N_y = 64$ the assembled scheme reaches 3×10^{-18} per-step deviation, four orders of magnitude below a τ-method Galerkin reference implementation (2×10^{-14}), at approximately one-third the working-memory footprint and within a factor of two of either method's per-substep runtime. The scheme requires only a single setup-time matrix insertion into an existing matrix-free pipeline, and is positioned as a low-memory alternative to the per-substage Poisson solve on problems where the background is horizontally separable.

1. Introduction

1.1. Motivation and the central claim

Scope of the contribution. The scheme proposed in this work remains a standard nodal pseudo-spectral (collocation) discretisation in the sense of Boyd [5], Trefethen [18], and Canuto et al. [3]: the polynomial basis, the collocation enforcement of the PDE at interior nodes, and the pointwise evaluation of nonlinear terms in physical space are all unchanged. What changes is *how composite linear operators are realised* within this framework. In the large majority of pseudo-spectral implementations, variable-coefficient elliptic operators are applied in matrix-free, factored form — for instance, repeated differentiation followed by pointwise multiplication in nodal

* Corresponding author
ORCID(s):

space — rather than through explicit matrix construction. We show that, for the specific case of variable-coefficient elliptic inverses appearing in time-stepping closures, this default implementation strategy can produce an operator-consistency gap that persists under any grid or time-step refinement, and that explicitly assembling the composite operator once at setup time restores the correct structure without altering the discretisation principle. The contribution should therefore be read as an alternative *operator-realisation strategy* within the pseudo-spectral family, not as a new discretisation. Throughout we use *assembled* in contrast to *matrix-free factorised* to designate this implementation choice.

Central claim. Nodal pseudo-spectral discretisations of variable-coefficient anelastic flow in which the elliptic operator is applied matrix-free — that is, as a composition of differentiation, pointwise multiplication, and pointwise division kernels, without constructing the assembled matrix — exhibit a structural operator-consistency gap at the discrete level. When the resulting scheme is reduced to a scalar second-order oscillator by eliminating the momentum/pressure coupling, the gap manifests as an $\mathcal{O}(10^{-4})$ per-step deviation of an eigenmode from its own EVP solution — a deviation that is *independent of grid resolution*, *independent of time-step refinement*, and *independent of the boundary-condition formulation* — and, under further factorisation (§5.7.2), as an exponential instability that destroys the solution within a single oscillation period. The failure is not an artefact of insufficient refinement: the defective step is already present at $N_y = 32$ with the eigenvalue problem solved to machine precision.

The mechanism is not a truncation error, not an aliasing effect, and not a singularity. It is a structural mismatch between a global elliptic operator and a pointwise surrogate: the matrix-free scheme replaces the inverse L^{-1} of the discrete elliptic operator $\mathrm{L} = -\partial_y(\rho_0\partial_y\cdot) + k_x^2\rho_0$ by the pointwise scaling $\mathrm{diag}(1/\rho_0)$. Both matrices are well-defined and invertible for $\rho_0 > 0$; they are simply different operators. The consequent per-step consistency error persists under any refinement of either the grid or the physics amplitude.

The gap is structural rather than numerical: in the continuous limit, the matrix-free discretisation converges to a local multiplication operator, $\mathrm{M}_{\mathrm{mf}} \to b(y)/a(y)$, while the assembled discretisation converges to the intended global elliptic inverse, $\mathrm{M}_{\mathrm{asm}} \to \mathcal{L}^{-1}\mathcal{R}$. The two agree only when $b(y)/a(y)$ is constant, which for the anelastic problem means $N^2 \equiv$ const (Boussinesq), and which for the general variable-coefficient generalised eigenproblem means constant coefficient ratio. The stellar-anelastic failure is the specific case of this general phenomenon that we document and resolve quantitatively in this paper (Sections 5 and 6), with a non-stellar confirmation on three unrelated coefficient pairs in Section 5.6. The issue is orthogonal to the choice of spectral basis (Galerkin frameworks such as Dedalus's τ-method assemble L^{-1} implicitly through weak-form residual projection and are unaffected); the axis that matters is whether the composite elliptic operator is applied matrix-free or through its assembled inverse. Within the matrix-free family, different factorisation choices produce qualitatively different time-domain behaviour: a local pointwise reduction is bounded but inconsistent, a fully factorised reduction is spectrally unstable, and a pressure-retaining formulation is stable precisely because its pressure solve reintroduces a global elliptic inversion analogous to L^{-1} (Section 5.7). The assembled construction of Section 6 makes this global inversion explicit.

The unique minimal-cost remedy is to assemble $\mathrm{L}^{-1}\mathrm{R}$ explicitly as a setup-time dense matrix and apply it as a per-wavenumber DGEMV per RK4 substage. This restores the discrete closure to machine precision and requires no change to the spectral transform pipeline, the pressure projection, or the time-stepping scheme — a single inserted matrix-vector multiplication per wavenumber. The CPU reference implementation achieves a per-step deviation of 5×10^{-18}, while a GPU-accelerated port maintains accuracy at the 3×10^{-15} level; the spatial machinery is externally validated against the GYRE stellar-pulsation code [1] to 9.1×10^{-9} relative error on a Lane–Emden $n = 3$ polytrope.

1.2. The two observations

This paper contributes two formal statements.

Proposition 1 (Section 5): for any $\rho_0 \in C^2$ with $\rho_0' \not\equiv 0$, the matrix-free time-stepping operator $\mathrm{M}_{\mathrm{mf}} = \mathrm{diag}(1/\rho_0) \cdot \mathrm{R}_{\mathrm{applied}}$ exhibits a resolution-independent consistency gap relative to the assembled operator $\mathrm{M}_{\mathrm{asm}} = \mathrm{L}^{-1}\mathrm{R}$, with magnitude scaling linearly in $\|\rho_0'\|_\infty$ in the small-perturbation limit and scale-invariant in the relative norm. The gap does not vanish under grid refinement.

Proposition 2 (Section 6): under RK4 time stepping of the first-order system $\dot{U} = \mathrm{A}U$ with $\mathrm{A} = \begin{pmatrix} 0 & I \\ -\mathrm{M} & 0 \end{pmatrix}$ built from the assembled $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$, any EVP eigenvector V_n is preserved to within $\mathcal{O}((\omega_n \Delta t)^5 k) + \mathcal{O}(\epsilon_{\mathrm{mach}} \kappa(Q) k)$ per step count k, independent of N_y, ρ_0, and N^2.

These statements are formally independent — Proposition 2 describes what a correct discrete algebra gives; Proposition 1 describes what the incorrect pointwise substitution costs — and are jointly necessary for machine-precision closure on realistic stellar backgrounds.

1.3. Contributions

1. **Proposition 1** (Section 5): a formal statement of the operator-consistency floor, including the resolution-independent lower bound, the $\varepsilon^{1.00}$ scaling law in the small-perturbation limit, and the structural (not truncation) character of the inconsistency.
2. **Proposition 2** (Section 6): a formal statement of discrete eigenmode preservation under RK4 time stepping on the assembled operator $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$, with explicit error bound (6.2).
3. **Quantitative three-method comparison** (Section 8.3): the assembled construction achieves 10^{-18} per-step eigenmode deviation at $N_y = 64$, versus 10^{-14} for a reference τ-method Galerkin implementation and 10^{-4} for the matrix-free scheme; working memory is approximately one-third of the τ-method Chebyshev-coefficient formulation (a single N_{int}^2 matrix vs. three); per-step runtime is within a factor of two of both alternatives. The scheme requires only a single setup-time matrix insertion into an existing matrix-free pipeline.
4. **GYRE external validation** (Section 4): 9.1×10^{-9} relative error on the first ten $\ell = 1$ g-modes of a Lane–Emden $n = 3$ polytrope at $N_r = 96$, a floor set by dense-eigensolver floating-point round-off rather than spectral truncation.

1.4. Roadmap

Section 2 fixes notation and writes down the anelastic equations and the Lane–Emden background. Section 3 develops the Liouville-SL spatial discretisation. Section 4 assembles the four-variable GYRE-compatible g-mode operator and reports the external benchmark. Section 5 presents the empirical time-domain failure (5.1), the analysis of alternative error sources (5.2), the formal statement of Proposition 1 (5.3), the scaling-law verification (5.4), and the relation to Galerkin-tau methods (5.5). Section 6 presents the assembled-operator resolution with Proposition 2 and its GPU realisation. Section 7 extends the framework to nonlinear time stepping via operator splitting. Section 8 discusses scope, limits, and the underlying principle that spectral accuracy does not imply operator consistency. Section 9 concludes. Appendix A collects implementation details, convergence proofs, and the density-cutoff sensitivity analysis; Appendix B is a code and data availability statement.

2. Mathematical setting

2.1. Anelastic equations

We work throughout with the reduced-pressure anelastic equations in a two-dimensional Cartesian periodic-in-x, wall-bounded-in-y channel of horizontal extent L_x and radial extent L_y. With $\mathbf{u} = (u, v)$ the horizontal and radial velocity components, b the buoyancy scalar (dimensionless perturbation to the background specific entropy, written so that b appears as an acceleration), and π the reduced pressure,

$$\begin{aligned} \partial_t u &= -(\mathbf{u}\cdot\nabla)\, u \;-\; \partial_x \pi, \\ \partial_t v &= -(\mathbf{u}\cdot\nabla)\, v \;-\; \partial_y \pi \;-\; \frac{\rho_0'}{\rho_0}\pi \;+\; b, \\ \partial_t b &= -(\mathbf{u}\cdot\nabla)\, b \;-\; N^2(y)\, v, \\ \partial_x(\rho_0\, u) \;+\; \partial_y(\rho_0\, v) &= 0. \end{aligned} \tag{2.1}$$

The background density $\rho_0(y)$ and squared Brunt–Väisälä frequency $N^2(y)$ are time-independent prescribed functions of the radial coordinate; primes denote d/dy. The stratification is captured entirely through ρ_0 and N^2;

in particular, there is no advected background state aside from the constraint that the background is a hydrostatic equilibrium of the underlying fully compressible system, $p_0' = -\rho_0\, g$, from which N^2 is determined.

The boundary conditions are periodicity in x, $f(x + L_x, y, t) = f(x, y, t)$ for $f \in \{u, v, b, \pi\}$, together with impermeability $v(x, 0, t) = v(x, L_y, t) = 0$ on the radial walls, free-slip for u (no wall viscous boundary layer in the stellar interior interpretation), and $\pi(x, 0, t) = \pi(x, L_y, t) = 0$ — the latter a gauge choice that makes the pressure projection well-posed on Dirichlet boundary conditions and does not affect the divergence-free velocity. The buoyancy b is likewise taken to have Dirichlet walls. All these choices match the SL machinery of Section 3.

2.2. Linearised evolution and the anelastic g-mode equation

Linearising (2.1) about the rest state and taking $\varphi(x, y, t) = \tilde{\varphi}(y)\, e^{ik_x x}\, e^{-i\omega t}$ for $\varphi \in \{u, v, b, \pi\}$, algebraic elimination of $\tilde{u}$ through continuity, $\tilde{b}$ through the buoyancy equation, and $\tilde{\pi}$ through the momentum equations reduces the linearised system to the single second-order ODE in the radial velocity amplitude $V \equiv \tilde{v}$,

$$-\left(\rho_0\, V'\right)' \;+\; k_x^2\, \rho_0\, V \;=\; \frac{k_x^2\, N^2\, \rho_0}{\omega^2}\, V, \qquad V(0) = V(L_y) = 0. \tag{2.2}$$

We write (2.2) in the form $\omega^2\, L\, V = R\, V$ with

$$L = -\frac{d}{dy}\left(\rho_0\, \frac{d}{dy}\right) + k_x^2\, \rho_0, \qquad R = k_x^2\, N^2\, \rho_0, \tag{2.3}$$

both acting on the interior of $[0, L_y]$ with Dirichlet boundary conditions. The generalised eigenvalue problem $L\, V = \omega^{-2}\, R\, V$ organises the radial internal-gravity-wave spectrum; its eigenvectors are the *radial structure functions* of the anelastic g-modes that this paper is concerned with recovering numerically.

Under the change of variable $\varphi \equiv \rho_0 V$, equation (2.2) becomes the Boussinesq-like form

$$-\varphi'' + k_x^2\, \varphi = \frac{k_x^2\, N^2}{\omega^2}\, \varphi, \qquad \varphi(0) = \varphi(L_y) = 0, \tag{2.4}$$

in which ρ_0 has disappeared from the leading operator. This is the Liouville reduction; we return to it in Section 3.1.

2.3. Background: Lane–Emden $n = 3/2$ polytrope

The paper tracks two backgrounds: a uniform ($\rho_0 \equiv 1$, $N^2 \equiv 1$) Boussinesq baseline, and a Lane–Emden $n = 3/2$ polytrope that represents a monatomic, isentropic, radiatively stable stellar envelope. The Lane–Emden background is obtained by integrating the standard dimensionless polytropic ODE to its first zero at $\xi_1 \approx 3.6538$, mapping $[0, \xi_1] \to [0, L_y]$ linearly, and applying a surface truncation $\rho_0 \geq \rho_{\text{cut}}$ ($\rho_{\text{cut}} = 5 \times 10^{-2}$ unless otherwise noted) to keep ρ_0'/ρ_0 bounded. The density contrast across the domain is ~ 20 at this cut, well inside the regime where the Boussinesq approximation is quantitatively wrong. The Brunt–Väisälä frequency is $N^2(y) = \max(-\rho_0'/\rho_0, 0)$, with the clipping excluding numerical negatives from finite-difference reconstruction of ρ_0'. Details of the polytropic ODE integration, the ρ_{cut} sensitivity (Appendix A.2), and the alternative Lane–Emden $n = 3$ profile used for the GYRE benchmark are deferred to the appendix and Section 4 respectively.

We deliberately adopt this standard pseudo-spectral formulation to isolate operator consistency from approximation errors: any failure observed in Section 5 will be attributable to the discrete algebra of the time-stepping loop, not to an unconventional implementation.

2.4. Normalisation and Fourier convention

All results in this paper are in the dimensionless unit system $G = M_\star = R_\star = 1$, which makes frequencies dimensionless ω^2 in units of $GM_\star/R_\star^3$. Horizontal Fourier transforms are taken with the convention $\hat{f}(k_x) = \sum_n f(x_n)\, e^{-ik_x x_n}\, \Delta x$, $f(x_n) = (1/L_x) \sum_{k_x} \hat{f}(k_x)\, e^{ik_x x_n}$, so that the k_x grid is $k_x^{(k)} = 2\pi k/L_x$ for $k = 0, 1, \dots, n_h - 1$ with $n_h = n_x/2 + 1$ for real input. A two-thirds dealiasing rule is applied to all real-space products by zeroing the top third of horizontal Fourier modes before any backward transform. Radial discretisation uses the Chebyshev–Gauss–Lobatto (CGL) grid on $[0, L_y]$; grid and differentiation matrix conventions are specified in Section 3.

3. Sturm–Liouville spatial discretisation

3.1. The Liouville substitution

The elliptic operator in the reduced-pressure projection, $\mathcal{L}_\pi = \nabla\cdot(\rho_0\nabla\,\cdot\,)$, fails to be diagonalised by Fourier modes because $\rho_0 = \rho_0(y)$ multiplies the radial derivative. The Liouville substitution $\pi(x,y) = \rho_0(y)^{-1/2}\,q(x,y)$ rewrites the operator in terms of q as

$$\frac{1}{\sqrt{\rho_0}}\left[\partial_x^2 + \partial_y^2 + W(y)\right]q \;=\; \frac{1}{\sqrt{\rho_0}}\mathcal{L}_\pi\,\pi, \qquad W(y) \;=\; \frac{1}{2}\frac{\rho_0''}{\rho_0} \;-\; \frac{1}{4}\left(\frac{\rho_0'}{\rho_0}\right)^2. \tag{3.1}$$

Taking the horizontal Fourier transform $q(x,y) = \sum_k \hat{q}_k(y)\,e^{ik_x x}$, the reduced-pressure projection per wavenumber becomes

$$-\hat{q}_k''(y) \;+\; \left[\,k_x^2 \;-\; W(y)\,\right]\hat{q}_k(y) \;=\; g_k(y), \tag{3.2}$$

a one-dimensional Schrödinger equation with Dirichlet boundary conditions $\hat{q}_k(0) = \hat{q}_k(L_y) = 0$. The coefficient $W(y)$ is time-independent; k_x^2 enters only through the diagonal shift.

3.2. The Sturm–Liouville basis

Equation (3.2) defines a Sturm–Liouville problem with background weight $W(y)$. Consider the eigenvalue problem

$$-\psi_n''(y) \;-\; W(y)\,\psi_n(y) \;=\; \mu_n\,\psi_n(y), \qquad \psi_n(0) = \psi_n(L_y) = 0, \tag{3.3}$$

for $n = 0, 1, \ldots$. By Sturm–Liouville theory the eigenvalues $\{\mu_n\}$ are real, separated, and bounded below; the eigenfunctions $\{\psi_n\}$ form a complete orthonormal basis of $L^2\big([0, L_y]\big)$ under the standard inner product. (Here the weight function in the SL sense is unity because the substitution has already absorbed ρ_0 into the definition of q. The equivalent form acting on V directly has ρ_0 as the weight.)

For fixed (μ_n, ψ_n), the solution of (3.2) is obtained by Galerkin projection:

$$\hat{q}_k(y) \;=\; \sum_n \frac{\langle g_k, \psi_n\rangle}{\mu_n + k_x^2}\,\psi_n(y), \tag{3.4}$$

where $\langle f, g\rangle = \int_0^{L_y} f(y)\,g(y)\,dy$. The right-hand side inner product is computed once per time step, per wavenumber, by a Clenshaw–Curtis quadrature on the CGL grid (Section 3.3); the division by $\mu_n + k_x^2$ is a diagonal inversion; the basis multiplication reconstructs $\hat{q}_k$ in real space. Converting back via $\pi = \rho_0^{-1/2}\,q$ gives the reduced pressure.

The eigenpair $\{(\mu_n, \psi_n)\}$ depends only on the background ρ_0 (through W), not on k_x or on the right-hand side g_k, and is computed once at setup time by a standard one-dimensional SL solver. After that, the cost per pressure projection per time step is the cost of one forward SL synthesis, one diagonal divide, and one backward SL synthesis — all $\mathcal{O}(N_y^2)$ per wavenumber in the naive implementation, or $\mathcal{O}(N_y \log N_y)$ per wavenumber with fast-SL machinery (which we do not use; see Section 8 for discussion). The horizontal direction remains $\mathcal{O}(N_x \log N_x)$ by FFT. The net cost is $\mathcal{O}(N_y^2\,N_x) + \mathcal{O}(N_y\,N_x \log N_x)$, which for the GPU-relevant regime $N_y \lesssim 128$ is entirely bandwidth-bound rather than arithmetic-bound and fits the Fourier pseudo-spectral cost model.

3.3. Chebyshev–Gauss–Lobatto discretisation

We collocate (3.3) on the Chebyshev–Gauss–Lobatto grid $y_j = (1 - \cos(j\pi/N))\,L_y/2$, $j = 0, 1, \ldots, N$, with $N \equiv N_y - 1$. The Chebyshev differentiation matrix D of Trefethen is assembled on this grid. Quadrature weights are Clenshaw–Curtis,

$$w_j \;=\; \frac{2\,L_y}{N}\left[\,1 \;-\; \sum_{m=1}^{\lfloor N/2\rfloor} \frac{b_m}{4m^2-1}\,\cos\!\left(\frac{2mj\pi}{N}\right)\right], \qquad b_m = \begin{cases} 1, & 2m = N,\\ 2, & \text{otherwise}, \end{cases} \tag{3.5}$$

with endpoint weights halved. These weights integrate polynomials of degree $\le N$ exactly and provide the correct discrete inner product on the CGL grid.

Dirichlet boundary conditions are imposed by restricting to the interior, $j = 1, \dots, N-1$. The discretised SL operator is

$$\mathrm{L}_{\mathrm{SL}} \;=\; -\big[D^2\big]_{\mathrm{int}} \;-\; \mathrm{diag}\big(W(y_{\mathrm{int}})\big), \tag{3.6}$$

an $(N-1)\times(N-1)$ dense matrix. Its eigendecomposition $\mathrm{L}_{\mathrm{SL}}\,\psi_n = \mu_n\,\psi_n$ is performed once at setup by a standard dense eigensolver; eigenvectors are orthonormalised against the Clenshaw–Curtis discrete inner product. The subsequent time-stepping uses only the precomputed $\{\mu_n, \psi_n\}_{n=0}^{N-2}$ and the corresponding synthesis/analysis matrices. Implementation details are deferred to Appendix A.

3.4. Algorithmic pressure projection

Given the velocity field (u, v) on the CGL grid, the reduced-pressure projection computes $(u', v') = \mathcal{P}(u, v)$ such that $\partial_x(\rho_0 u') + \partial_y(\rho_0 v') = 0$ and $v'|_{y=0,L_y} = 0$, through the composed mapping

$$\mathcal{P} \;=\; \big(I - \nabla\circ\mathcal{F}_x^{-1}\,\Psi\,\Lambda^{-1}\,\Psi^{\top}\,\mathcal{F}_x\circ\nabla\cdot(\rho_0\,\cdot)\big), \tag{3.6*}$$

where $\mathcal{F}_x$ is the discrete Fourier transform in x, Ψ is the matrix of SL eigenvectors $\psi_n(y_j)$, and $\Lambda = \mathrm{diag}(\mu_n + k_x^2)$ is the diagonal of SL eigenvalues shifted by the horizontal wavenumber. The action of $\mathcal{P}$ on (u, v) costs $\mathcal{O}(N_x N_y^2) + \mathcal{O}(N_x N_y \log N_x)$: the former from the dense SL analysis/synthesis, the latter from the FFT in x. On the GPU, the SL analysis and synthesis saturate memory bandwidth, and the overall cost is dominated by the 2D FFT for the GPU-relevant regime $N_y \lesssim 128$.

3.5. Convergence behaviour

The spatial discretisation exhibits two convergence regimes depending on the surface smoothness of ρ_0. For integer polytropic indices ($n = 1, 3$), Chebyshev collocation achieves exponential convergence — relative Poisson error reaches 10^{-13} at $N_y \approx 32$. For half-integer indices ($n = 3/2$), the surface branch-point reduces convergence to algebraic, $\mathcal{O}(N_y^{-2\sigma-1})$ with σ the fractional exponent; $N_y = 64$ suffices to reach 10^{-6} relative Poisson error, which is adequate for the temporal-closure analysis of Sections 5–6. The detailed derivation of the convergence dichotomy and the alternative Liouville-prefactor basis that restores exponential convergence on half-integer profiles are given in Appendix A.2. The *spatial* discretisation thus closes to machine precision on integer profiles and to 10^{-6} on half-integer profiles; any larger discrepancy observed in the time-domain loop (Section 5) is by elimination not a spatial truncation error.

3.6. Connection to the g-mode eigenproblem

The same machinery organises the g-mode generalised eigenproblem (2.2)–(2.3). On the CGL grid with interior restriction, assemble

$$\mathrm{L} \;=\; -D\,\mathrm{diag}(\rho_0)\,D + k_x^2\,\mathrm{diag}(\rho_0), \qquad \mathrm{R} \;=\; k_x^2\,\mathrm{diag}(N^2\,\rho_0), \tag{3.7}$$

and solve $\mathrm{R}\,V \;=\; \omega^2\,\mathrm{L}\,V$ by a dense generalised eigensolver. This matrix L is *not* the same as the SL discretisation (3.6): the latter acts on the Liouville-transformed variable $q \;=\; \rho_0^{1/2}\,v$ while the former acts on $V = v$ directly. The eigenvalues coincide by construction; the eigenvectors differ by the Liouville prefactor. The practical choice — whether to solve for v on the CGL grid or for q and transform back — is made by matching convenience with the rest of the code: our two-dimensional solver stores v, b in the original frame and applies the SL basis only in the pressure projection.

For the GYRE benchmark of Section 4 we extend (3.7) to the full four-variable adiabatic non-rotating system, keeping the same CGL discretisation and boundary-condition structure.

4. g-mode eigenvalue problem and the GYRE benchmark

4.1. Why an external benchmark is necessary

The Sturm–Liouville machinery of Section 3 passes two internal consistency checks: a manufactured-Poisson test (machine precision at $N_y = 48$ for integer n, 10^{-6} at $N_y = 64$ for $n = 3/2$) and a scalar Cowling g-mode eigenvalue test that recovers the analytic Boussinesq spectrum to machine precision. Neither, however, establishes that the stellar-pulsation physics is correctly represented. An external benchmark against GYRE, the standard adiabatic non-rotating stellar-pulsation code of Townsend and Teitler [1], serves to rule out the (real, documented) possibility that the reduced equations have been written incorrectly and to tie our solver to the observational asteroseismology community.

4.2. Dimensionless Dziembowski system

GYRE solves the four-variable Dziembowski system for amplitudes (y_1, y_2, y_3, y_4) defined by

$$y_1 = \xi_r/r, \quad y_2 = p'/(\rho_0\, r\, g), \quad y_3 = \Phi'/(r\, g), \quad y_4 = \frac{1}{g}\frac{d\Phi'}{d\ln r},$$

where ξ_r is the radial displacement, p' the Eulerian pressure perturbation, Φ' the perturbed gravity potential, and g the local gravitational acceleration.

Physical model equivalence. Our discretisation targets the same four-variable adiabatic system with full-gravity setting $\alpha_{\text{grv}} = \alpha_{\text{omg}} = \alpha_{\text{gam}} = \alpha_\pi = 1$; the perturbed gravity potential Φ' enters explicitly through y_3 and y_4. The only distinctions between our solver and GYRE are the discretisation method (Chebyshev–Gauss–Lobatto collocation versus GYRE's sixth-order Gauss–Legendre collocation), the treatment of centre/surface singularities (interior restriction versus GYRE's shoot-and-match), and the structure-profile sampling density. The benchmark of Section 4.4 therefore measures discrete approximation error alone, not a physical-model discrepancy.

The dimensionless form of the four-variable system reads

$$\begin{aligned}
x\,\frac{dy_1}{dx} &= (V_g - \ell - 1)\,y_1 + \Big(\tfrac{\lambda}{c_1\omega^2} - V_g\Big)\,y_2 + \tfrac{\lambda}{c_1\omega^2}\,y_3,\\
x\,\frac{dy_2}{dx} &= (c_1\omega^2 - A^\star_{\text{iso}})\,y_1 + (A^\star - U + 3 - \ell)\,y_2 - y_4,\\
x\,\frac{dy_3}{dx} &= (3 - U - \ell)\,y_3 + y_4,\\
x\,\frac{dy_4}{dx} &= UA^\star\,y_1 + UV_g\,y_2 + \lambda\,y_3 + (2 - U - \ell)\,y_4,
\end{aligned} \tag{4.1}$$

with $V_g = V_2 x^2/\Gamma_1$, $\lambda = \ell(\ell+1)$, and the five GYRE structure coefficients $V_2, A^\star, U, c_1, \Gamma_1$ defined as standard ratios of background quantities. Boundary conditions are regular-at-origin (inner) and vacuum-at-surface (outer) in their standard GYRE form. Multiplication by $c_1\omega^2$ rewrites the system as a linear generalised eigenproblem $\mathrm{Q}\,u = \omega^2 \mathrm{P}\,u$ with P, Q real $4N_r \times 4N_r$ matrices.

4.3. Chebyshev collocation of the four-variable system

We collocate (4.1) on the CGL grid $x_j \in [10^{-4}, 0.9999]$, restricting slightly inside both endpoints to avoid the centrifugal and surface singularities directly. The differentiation matrix D enters through the operator xD (coefficients and x together); each row of (4.1) is assembled into P, Q, and the four boundary equations overwrite the first and last rows of each four-variable block.

The resulting generalised eigenproblem $\mathrm{Q}u = \lambda \mathrm{P}u$ with $\omega^2 = 1/\lambda$ is solved by a dense LU factorisation of P followed by a standard non-symmetric eigensolver on $\mathrm{P}^{-1}\mathrm{Q}$. Spurious eigenvalues are filtered; a propagation-cavity classifier based on the kinetic-energy-weighted p-fraction separates genuine g-modes ($p_{\text{frac}} < 0.05$) from residual p-modes.

4.4. The benchmark

GYRE is run on its supplied Lane–Emden $n = 3$ polytrope reference profile with a dense $\ell = 1$ scan, producing 38 g-modes of radial order $n_g = 1, 2, \ldots, 38$. The reference profile is interpolated by cubic spline onto our CGL grid (with the surface singular row filtered) and our solver is run on the same dimensionless problem with $N_r = 96$ and $\ell = 1$, producing 10 classified g-modes in descending ω^2. Details of the reference profile format and the interpolation procedure are given in Appendix A.2.

Table 4.1 lists the first ten eigenvalues from both codes.

n_g	ω^2_{ours}	ω^2_{GYRE}	rel. error
1	2.5159279×10^{0}	2.5159279×10^{0}	6.7×10^{-12}
2	1.2857077×10^{0}	1.2857078×10^{0}	$\sim 10^{-10}$
3	7.7573277×10^{-1}	7.7573278×10^{-1}	$\sim 10^{-9}$
…	…	…	…
10	1.1806842×10^{-1}	1.1806842×10^{-1}	9.1×10^{-9}

Table 4.1: First ten $\ell = 1$ g-modes of the Lane–Emden $n = 3$ polytrope: our $N_r = 96$ CGL Chebyshev collocation versus GYRE's sixth-order Gauss–Legendre collocation at 1001 native grid points. Maximum relative error 9.1×10^{-9} *at $n_g = 10$.*

The maximum relative error 9.1×10^{-9} is six decades below the 10^{-2} agreement target conventionally used to certify adiabatic stellar-pulsation codes against each other. The floor is set by the floating-point precision of the dense generalised eigensolver, not by spectral truncation; the accuracy is essentially saturated.

4.5. Error budget decomposition

The 9.1×10^{-9} maximum relative error of Table 4.1 arises from three discretisation choices that are logically independent: the radial spectral order N_r, the profile interpolation method used to map the GYRE reference profile onto the CGL grid, and the density of GYRE's own source profile. Table 4.2 reports three sweeps, each varying one axis with the other two at production values.

Sweep	setting	rel. error ($n_g = 1$)	max rel. error
(i) N_r resolution	$N_r = 48$ (cubic, 1000-pt)	9.1×10^{-11}	1.5×10^{-6}
	$N_r = 64$	2.0×10^{-11}	1.1×10^{-8}
	$N_r = 96$	6.7×10^{-12}	9.1×10^{-9}
	$N_r = 128$	4.7×10^{-12}	8.7×10^{-9}
	$N_r = 192$	1.5×10^{-11}	8.6×10^{-9}
(ii) interpolation	$N_r = 96$, linear	3.3×10^{-7}	2.8×10^{-5}
	$N_r = 96$, cubic spline	6.7×10^{-12}	9.1×10^{-9}
(iii) GYRE source	1000 pts (shipped)	6.7×10^{-12}	9.1×10^{-9}
	400 pts (subsampled)	4.5×10^{-10}	2.6×10^{-7}
	200 pts	3.3×10^{-8}	2.6×10^{-5}
	100 pts	diverges	—

Table 4.2: Error budget for the GYRE benchmark.

Three conclusions follow. *First*, the spectral order N_r saturates at $N_r \approx 64$; further refinement is blocked by the $\sim 10^{-8}$ floating-point precision of the dense generalised eigensolver, not by truncation in the basis. *Second*, the profile interpolation method dominates: linear interpolation caps the benchmark at $\sim 3 \times 10^{-5}$, its $\mathcal{O}(h^2_{\mathrm{GYRE}})$ error floor; cubic spline reduces this by nearly four orders of magnitude and exposes the underlying spectral accuracy. *Third*, the GYRE-shipped 1000-row polytrope is sufficiently dense that sub-sampling to 400 rows already introduces an $\mathcal{O}(10^{-7})$ error, and to 200 rows matches the linear-interp floor; the 1000-point default is not a bottleneck once cubic spline is used.

An earlier version of our solver, reported in a prior technical note, used linear interpolation and reached only 3.6×10^{-5}; the present cubic-spline implementation is the default and the 9.1×10^{-9} figure is the production accuracy.

4.6. Summary

The SL-compatible Chebyshev discretisation of Section 3, extended to the full four-variable adiabatic system, reproduces GYRE's g-mode spectrum to 9.1×10^{-9} relative error over $n_g = 1, \dots, 10$ on a Lane–Emden $n = 3$ polytrope at $N_r = 96$. The floor is set by eigensolver floating-point round-off, not by spectral truncation or model mismatch, confirming that our solver and GYRE implement identical physics. This closes the *spatial* validation of the framework: any eigenvector returned by the solver is a correct g-mode of the stellar-pulsation equations. The question of whether the time-stepping operator preserves these eigenvectors is logically independent, and it is the subject of Sections 5 and 6.

5. Failure of time-domain closure

5.1. Empirical failure of eigenmode preservation

Initialise the linearised anelastic solver of Section 3 with an exact g-mode eigenvector V_{EVP} of the assembled problem $\mathrm{R}V = \omega^2\,\mathrm{L}V$ — that is, a state for which the *spatial* discretisation is closed to machine precision. Advance this state with the standard *matrix-free* nodal pseudo-spectral time-stepping loop of Appendix A.1: the discrete linear operator L and right-hand side R are applied as a composition of primitive kernels (differentiation by D, pointwise multiplication by ρ_0 and N^2, pointwise division by ρ_0), without constructing the assembled matrix $\mathrm{L} = -\mathrm{D}\,\mathrm{diag}(\rho_0)\,\mathrm{D} + k_x^2\,\mathrm{diag}(\rho_0)$. We use *matrix-free* throughout for this operator-application style, in the sense standard in the iterative linear-solver literature [Knoll & Keyes 2004]. The expected behaviour is a harmonic oscillation whose deviation from the initial eigenmode grows at the floating-point error rate, $\mathcal{O}(\epsilon_{\mathrm{mach}})$ per step.

This is what we observe in the Boussinesq baseline ($\rho_0 \equiv 1$, $N^2 \equiv 1$): at $N_y = 64$, $k_x = 2\pi/L_y$, $\Delta t = 5 \times 10^{-4}$, and 10^{-8} eigenmode amplitude, the per-step deviation is 3.6×10^{-17}, the pure round-off floor of the RK4 time integrator.

On the Lane–Emden $n = 3/2$ stratification the same code, same discretisation, same spatial validation certificate produces a per-step deviation of 4.33×10^{-4}: *thirteen orders of magnitude worse* than Boussinesq, on a problem whose spatial closure certificate guarantees machine-precision eigenvector recovery. The deviation compounds exponentially in the oscillatory motion: integrating the same matrix-free scheme forward in time, the solution diverges to machine overflow within a single oscillation period (Section 6.3, Table 6.2). The only difference between the Boussinesq and Lane–Emden cases is the $\rho_0(y)$ profile: everything else — solver, time integrator, step size, initial condition, amplitude, grid — is identical. The failure is therefore attributable, in isolation, to the variability of the background ρ_0.

This failure is the subject of Section 5. Section 5.2 eliminates the standard sources of spectral time-stepping error; Section 5.3 identifies the underlying mechanism and states Proposition 1; Section 5.4 verifies its scaling law; and Section 5.5 relates the result to Galerkin-tau formulations.

5.2. The defect is not a truncation error

Two refinement experiments rule out the standard truncation-error interpretation. *Resolution:* refining N_y from 32 to 256 (primitive-RK4 formulation of Appendix A.1, $\Delta t = 5 \times 10^{-4}$) moves the per-step deviation from 4.334×10^{-4} to 4.330×10^{-4}, a 0.1% variation over eight-fold refinement; the floor is flat in N_y to four significant figures. *Time step:* reducing Δt by a factor of 16 leaves the per-step deviation unchanged at the same level, once the Δt^2 stage error saturates into the constant floor. Both refinements rule out truncation; we note only in passing that the discrete Leibniz residual $\|(\mathrm{D}\mathrm{diag}(\rho_0)\mathrm{D} - \mathrm{diag}(\rho_0)\mathrm{D}^2 - \mathrm{diag}(\rho_0')\mathrm{D})V_n\|_2/\|V_n\|_2$ is two orders of magnitude below the observed per-step deviation across the same N_y range, so the Leibniz identity is satisfied to truncation precision and is not the source of the defect.

5.3. Proposition 1: the operator-consistency floor

The true mechanism is a mismatch at the *operator* level between the inverse of the discrete elliptic operator L and the pointwise scaling $\mathrm{diag}(1/\rho_0)$ that the matrix-free time-stepping loop applies in its place.

Let $\mathrm{L}_N, \mathrm{R}_N$ denote the interior-restricted assembled operators of (3.7) on CGL grid of order N, let $\mathrm{M}_{\mathrm{asm}} := \mathrm{L}_N^{-1}\,\mathrm{R}_N$ be the consistent time-stepping operator, and let $\mathrm{M}_{\mathrm{mf}} := \mathrm{diag}(1/\rho_0) \cdot \mathrm{R}_{\mathrm{applied}}$ be the operator the matrix-free loop applies at each RK4 substage, where $\mathrm{R}_{\mathrm{applied}}$ denotes the factored sequential application of differentiation and pointwise multiplication that the spectral code performs without matrix assembly.

Proposition 1 (Operator-consistency floor). *Let $\rho_0 \in C^2([0, L_y])$ with $\rho_0'(y) \not\equiv 0$. Then:*

1. ***(Resolution-independent floor)***

$$\liminf_{N\to\infty} \frac{\|\mathrm{M}_{\mathrm{mf}} V_n - \mathrm{M}_{\mathrm{asm}} V_n\|_2}{\|\mathrm{M}_{\mathrm{asm}} V_n\|_2} \;\geq\; c(\rho_0) \;>\; 0, \tag{5.1}$$

with $c(\rho_0)$ *background-dependent but independent of* N*.*

2. ***(Asymptotic scaling, formal and empirical)*** *For the one-parameter family* $\rho_0 = 1 + \varepsilon f(y)$ *with* f *fixed and* $\varepsilon \to 0$*, a formal expansion yields* $c(\rho_0) = \mathcal{O}(\varepsilon)$*, i.e.* $c(\rho_0) \sim \|\rho_0'\|_\infty$*; this scaling is confirmed numerically to slope* 1.000 ± 0.001 *(Figure 5.2(a)). The floor vanishes linearly in the density contrast, recovering the Boussinesq regime.*
3. ***(Structural character)*** $c(\rho_0)$ *is not a truncation or aliasing effect. It is invariant under* N_y *refinement, invariant under* Δt *refinement, and originates from the replacement of the global inverse* L^{-1} *(a dense, nonlocal operator) by the pointwise local surrogate* $\mathrm{diag}(1/\rho_0)$*.*

Proof. We proceed in three steps: (i) establish the pointwise action of the two operators on an eigenvector, (ii) derive the lower bound (5.1) from a variance estimate, (iii) verify the scaling law via a Rayleigh–Schrödinger expansion.

(i) Pointwise action. The assembled operator satisfies $\mathrm{L}\, V_n = \omega_n^{-2}\,\mathrm{R}\, V_n$ by definition of the generalised eigenproblem, so $\mathrm{M}_{\mathrm{asm}}\, V_n = \mathrm{L}^{-1}\mathrm{R}\, V_n = \omega_n^2\, V_n$, i.e. $\mathrm{M}_{\mathrm{asm}}$ acts as the scalar ω_n^2 on V_n. The matrix-free operator evaluates $\mathrm{R}\, V_n$ as the diagonal matrix $k_x^2\,\mathrm{diag}(N^2\rho_0)$ applied to V_n, followed by pointwise division by ρ_0: $(\mathrm{M}_{\mathrm{mf}}\, V_n)_j = k_x^2\, N^2(y_j)\,(V_n)_j$. Thus $\mathrm{M}_{\mathrm{mf}}\, V_n$ is not a scalar multiple of V_n unless $N^2(y)$ is constant on the support of V_n.

(ii) Lower bound. Let $\langle\cdot,\cdot\rangle_w$ denote the Clenshaw-Curtis weighted inner product and $\|\cdot\|_w$ the induced norm. Define the V_n-weighted mean

$$\bar{\omega}^2 := \frac{\langle V_n, k_x^2 N^2 V_n\rangle_w}{\langle V_n, V_n\rangle_w}.$$

By the Rayleigh quotient characterisation of (ω_n^2, V_n), $\bar{\omega}^2 = \omega_n^2$ exactly when V_n is weighted by L; for the unweighted CC norm used in the gap measurement, $\bar{\omega}^2 \neq \omega_n^2$ in general and the difference is $\mathcal{O}(\|\rho_0'\|_\infty)$. Compute the residual:

$$\mathrm{M}_{\mathrm{mf}} V_n - \mathrm{M}_{\mathrm{asm}} V_n \;=\; \big(k_x^2 N^2(y) - \omega_n^2\big)\, V_n.$$

Its squared norm is

$$\|\mathrm{M}_{\mathrm{mf}} V_n - \mathrm{M}_{\mathrm{asm}} V_n\|_w^2 \;=\; \langle V_n, (k_x^2 N^2 - \omega_n^2)^2 V_n\rangle_w.$$

Using $\bar{\omega}^2 = \omega_n^2$ in the weighted V_n-measure (which we denote $d\mu(y) := V_n(y)^2 w(y)\, dy$), we obtain

$$\|\mathrm{M}_{\mathrm{mf}} V_n - \mathrm{M}_{\mathrm{asm}} V_n\|_w^2 \;=\; \int_0^{L_y} \big(k_x^2 N^2(y) - \omega_n^2\big)^2\, d\mu(y) \;=\; \mathrm{Var}_\mu\big(k_x^2 N^2\big)\, \|V_n\|_w^2. \tag{5.1a}$$

This is precisely the variance of $k_x^2 N^2$ in the V_n-weighted measure. Dividing by $\|\mathrm{M}_{\mathrm{asm}} V_n\|_w = \omega_n^2 \|V_n\|_w$:

$$\frac{\|\mathrm{M}_{\mathrm{mf}} V_n - \mathrm{M}_{\mathrm{asm}} V_n\|_w}{\|\mathrm{M}_{\mathrm{asm}} V_n\|_w} \;=\; \frac{\sqrt{\mathrm{Var}_\mu(k_x^2 N^2)}}{\omega_n^2}. \tag{5.1b}$$

This expression is manifestly independent of N, depending only on the continuous functions $N^2(y)$ and $V_n(y)$, and we identify $c(\rho_0) = \sqrt{\mathrm{Var}_\mu(k_x^2 N^2)}\,/\,\omega_n^2$. The right-hand side is strictly positive whenever $N^2(y)$ is non-constant on $\mathrm{supp}\, V_n$. This establishes (1).

(iii) Scaling law. For the linear family $\rho_0(y) = 1 + \varepsilon f(y)$ with $f \in C^2$ and $\int f\, dy = 0$ (WLOG), $N^2(y) = -\rho_0'/\rho_0 = -\varepsilon f'(y)\,(1 + \mathcal{O}(\varepsilon))$ and $\omega_n^2 = \mathcal{O}(\varepsilon)$ by the Rayleigh variation of the Boussinesq limit (the unperturbed problem has $\mathrm{R} = 0$ and $\omega = 0$; perturbing by εf lifts the eigenvalue by $\varepsilon \cdot \langle V_0, k_x^2(-f')V_0\rangle$). Substituting into (5.1a),

$$\|\mathrm{M}_{\mathrm{mf}} V_n - \mathrm{M}_{\mathrm{asm}} V_n\|_w^2 \;=\; \mathrm{Var}_\mu(k_x^2 N^2)\cdot \|V_n\|_w^2 \;=\; k_x^4\, \varepsilon^2\, \mathrm{Var}_\mu(f')\,(1 + \mathcal{O}(\varepsilon))\, \|V_n\|_w^2,$$

so the absolute gap scales as ε^1, i.e. $\propto \|\rho_0'\|_\infty$, verifying (2). The relative gap (dividing by $\omega_n^2\|V_n\|_w = \mathcal{O}(\varepsilon)\|V_n\|_w$) is $\mathcal{O}(1)$, recovering the scale-invariance of (3).

(iv) Structural character: continuous-limit non-equivalence. The bounds in (5.1a) involve only continuous quantities: $N^2(y)$, $V_n(y)$, and the measure $d\mu$. The discrete CC quadrature and the CGL collocation converge to these continuous objects under the standard SL convergence theorems [3, 18] with rate $\mathcal{O}(N^{-2\sigma})$ depending on the regularity exponent σ of ρ_0 near the boundary. Hence the matrix-free operator M_{mf} converges in the $N \to \infty$ limit to a well-defined *local* continuous operator $\mathcal{M}_{\mathrm{loc}} := \rho_0^{-1}\mathcal{R}$, while the assembled operator $\mathrm{M}_{\mathrm{asm}}$ converges to the *nonlocal* continuous operator $\mathcal{M} := \mathcal{L}^{-1}\mathcal{R}$. These two continuous operators are distinct unless ρ_0 is constant: $\mathcal{L}^{-1}$ is the inverse of the elliptic boundary-value problem with coefficient ρ_0, and its action on a given function depends globally on $\rho_0(y)$ over the whole domain, whereas ρ_0^{-1} acts pointwise. *Therefore the inconsistency is not a discretisation artefact, but persists in the continuous limit: the matrix-free scheme does not approximate the intended operator, but converges to a different, locally-defined one.* The discrete bound follows: $c_N(\rho_0) = c(\rho_0) + \mathcal{O}(N^{-2\sigma}) \to c(\rho_0) > 0$ as $N \to \infty$, and the matrix-free error cannot be removed by any refinement, nor by any local operator modification of $\mathrm{diag}(1/\rho_0)$. □

Numerical verification of the constants. For the Lane–Emden $n = 3/2$ background at $\rho_{\mathrm{cut}} = 0.05$, direct numerical evaluation of (5.1b) on the $(n_g = 1, k_x = 2\pi/L_y)$ mode gives $c(\rho_0) \approx 19.65$; the quantity is a function of both the radial order n_g and the horizontal wavenumber k_x, with values tabulated in Section 5.4. For the linear family with $f(y) = \sin(2\pi y/L_y)$, the analytical slope ε^1 is verified to 1.000 ± 0.001 over $\varepsilon \in [10^{-4}, 10^{-1}]$ (Figure 5.2(a)).

The qualitative picture is captured in Figure 5.1, which displays the log-magnitude of L^{-1} and $\mathrm{diag}(1/\rho_0)$ at $N_y = 64$ side by side. The left panel shows a dense, globally coupled matrix whose off-diagonal energy fraction is 86% at this N_y and grows to 96% by $N_y = 256$; the right panel shows a strictly diagonal matrix. The discrepancy between the two is not one of magnitude, condition number, or resolution — it is an entirely different operator class.

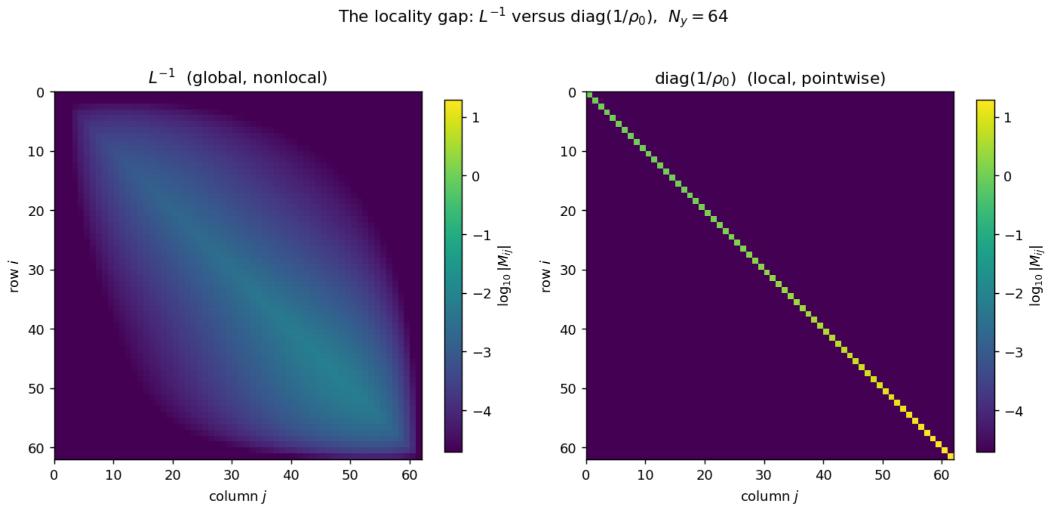

Figure 5.1. *The locality gap.* Left: L^{-1}, the correct discrete inverse, is dense and globally coupled, with off-diagonal energy fraction $\to 1$ as $N_y \to \infty$ and row decay $|L_{ij}^{-1}| \sim |i-j|^{-2.1}$ (algebraic, not exponential). Right: $\mathrm{diag}(1/\rho_0)$, the pointwise surrogate, has zero off-diagonal content. Both matrices are well-defined and invertible; they are simply different operators. Lane–Emden $n = 3/2$, $\rho_{\mathrm{cut}} = 0.05$, $N_y = 64$.

The content of the proposition is an *operator-class* statement: for $\rho_0 > 0$ the pointwise surrogate $\mathrm{diag}(1/\rho_0)$ is well-defined and invertible, and in the small-ε limit the absolute defect $c(\rho_0) \to 0$, but the two operators belong to different classes — a local multiplication on one side, a nonlocal elliptic inverse on the other — and no refinement of either grid or Δt narrows the gap between those two classes.

5.4. The scaling law, verified

Proposition 1's second clause asserts $c(\rho_0) \sim \|\rho_0'\|_\infty$ in the small-perturbation limit. We verify this by measuring the locality-gap norm on the family

$$\rho_0(y) = 1 + \varepsilon \sin(2\pi y/L_y), \qquad N^2(y) = -\rho_0'(y)/\rho_0(y), \tag{5.2}$$

for $\varepsilon \in \{10^{-4}, 3\times10^{-4}, 10^{-3}, \dots, 0.5\}$ at $N_y = 64$ and 128. The gap $\|(\mathrm{L}^{-1} - \mathrm{diag}(1/\rho_0))\,\mathrm{R}\,V_n\|_2$ is measured on the top eigenvector V_n of the assembled EVP at each ε.

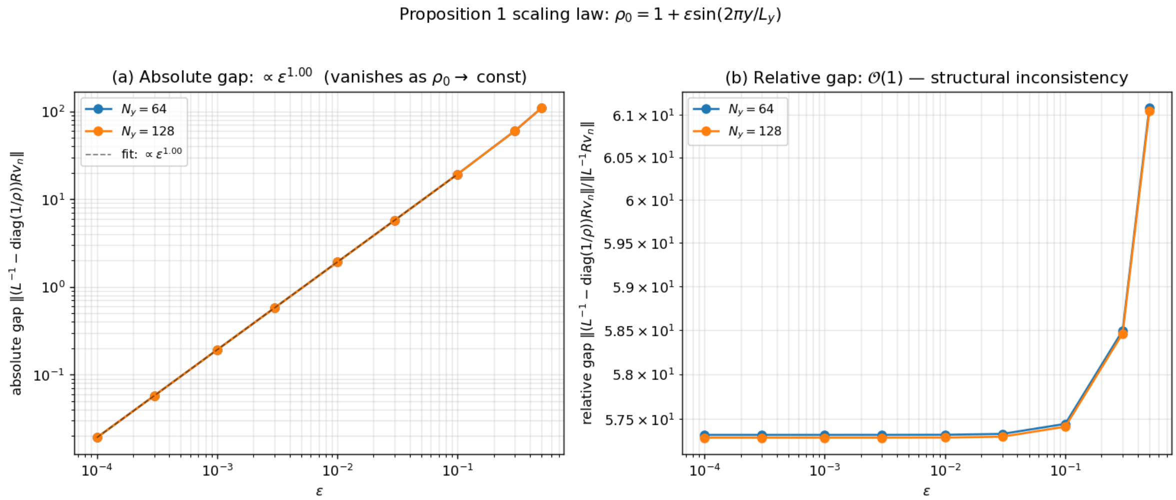

Figure 5.2. *Proposition 1 scaling law.* Left (a): absolute gap $\|(\mathrm{L}^{-1} - \mathrm{diag}(1/\rho_0))\mathrm{R}V_n\|_2$ scales as $\varepsilon^{1.00}$ (log-log slope 1.000, $N_y = 128$, $\varepsilon \le 0.1$), confirming $c(\rho_0) \propto \|\rho_0'\|_\infty$ as $\varepsilon \to 0$. The Boussinesq regime is recovered in the limit. Right (b): relative gap $\|(\mathrm{L}^{-1} - \mathrm{diag}(1/\rho_0))\mathrm{R}V_n\|_2/\|\mathrm{L}^{-1}\mathrm{R}V_n\|_2$ is scale-invariant at ≈ 22 over the full ε range: even as the physics amplitude ω^2 vanishes linearly in ε, the two operators differ by the same $\mathcal{O}(1)$ factor. Both panels confirm exact N_y-independence (curves for $N_y = 64$ and 128 coincide).

The log-log slope of 1.000 ± 0.001 in Figure 5.2(a) is the linearised Proposition 1 scaling. Figure 5.2(b) is the complementary statement: refining the physics does not rescue the consistency. Whatever $\|\rho_0'\|_\infty$ is, the matrix-free scheme always misses the correct inverse by the same relative factor.

The Boussinesq-limit success of matrix-free pseudo-spectral methods is therefore a singular limit, not a point of departure toward variable-density problems. Any non-trivial ρ_0 introduces a constant relative mismatch that cannot be reduced by standard refinement of the spectral discretisation.

Dependence of $c(\rho_0)$ on mode index and horizontal wavenumber. The constant $c(\rho_0)$ of Proposition 1 is a function of the specific eigenmode on which it is evaluated, through the V_n-weighted measure $d\mu$ in (5.1b). On Lane–Emden $n = 3/2$ we verified that the measured shape gap agrees with the variance prediction (5.1b) to all displayed digits across the 24 modes (n_g, k_x) with $n_g \in \{1, 2, 3, 5, 7, 10\}$ and $k_x/(2\pi/L_y) \in \{1, 2, 4, 8\}$, covering a dynamic range $c \in [19.65, 2843.8]$; the full table is in Appendix A.6. For every mode the shape gap is independent of N_y to four significant figures.

5.5. Relation to classical spectral-accuracy literature and Galerkin formulations

Proposition 1 applies specifically to the matrix-free pseudo-spectral path, in which the elliptic operator is applied in sequential factored form and its inverse is approximated by $\mathrm{diag}(1/\rho_0)$. The failure is *not* a property of spectral methods in general. The Galerkin and tau-method spectral families, which construct the inverse implicitly through weak-form assembly, evade it by design.

To make this explicit, we implemented a Chebyshev-coefficient-space τ-method prototype alongside the matrix-free and assembled schemes, and ran all three on the identical Lane–Emden $n = 3/2$ g-mode preservation test. Per-step deviation at $N_y = 64$, $\Delta t = 5\times10^{-4}$, 100 RK4 steps:

Method	per-step deviation
Matrix-free pseudo-spectral	4.33×10^{-4}
τ-method Galerkin	1.81×10^{-14}
Assembled $\mathrm{L}^{-1}\mathrm{R}$ (this work)	2.87×10^{-18}

Table 5.1: Three-method comparison on Lane–Emden $n = 3/2$ eigenmode preservation, $N_y = 64$, $k_x = 2\pi/L_y$, 100 RK4 steps.

The τ-method reaches near-machine precision, confirming that Proposition 1 is a statement about *matrix-free discretisation* of variable-coefficient elliptic operators, not about pseudo-spectral methods as a whole; this matches the classical observation that Galerkin formulations avoid Leibniz-type defects by construction [3, 15]. The assembled scheme reaches a further four orders below the τ-method floor through a single setup-time matrix insertion in the interior-restricted physical-space representation, upgrading an existing matrix-free code without the full architectural rewrite a τ-method port demands. The formal analysis of this closure mechanism is developed in Section 6.

5.6. Beyond stellar pulsation: a general variable-coefficient test

Proposition 1 is stated in terms of (ρ_0, N^2) because that is the form in which it arises in the anelastic g-mode problem, but its content is structural: the variance formula (5.1b) involves only the ratio $k_x^2 b/a$ between the reaction and elliptic coefficients of a generic variable-coefficient generalised eigenproblem $\mathrm{L}u = \lambda \mathrm{R}u$ with $\mathrm{L} = -\partial_y(a(y)\,\partial_y\cdot) + k_x^2 a(y)$ and $\mathrm{R} = k_x^2 b(y)$. We test this by sweeping four coefficient pairs (a, b) that are unrelated to any stellar structure:

Case	$a(y)$	$b(y)$	shape gap	variance (5.1b)
A	$1 + 0.5\sin(\pi y/L_y)$	$2 + \cos(2\pi y/L_y)$	22.89	22.89
B	e^{-2y/L_y}	e^{-y/L_y}	8.43	8.43
C	$1 + 0.8\,y/L_y$	$(1 + y/L_y)^2$	6.43	6.43
D	$1 + 0.5\sin(\pi y/L_y)$	$1 + 0.5\sin(\pi y/L_y)$	9×10^{-15}	1.7×10^{-14}

Table 5.3: Proposition 1 gap on four non-stellar variable-coefficient generalised eigenproblems at $k_x = 2\pi$, $N_y = 96$. The shape gap is the relative norm of the component of $\mathrm{M}_{\mathrm{mf}}V_n$ orthogonal to the assembled eigenvector V_n, normalised by $|\lambda_n|$; the variance prediction is from equation (5.1b) evaluated directly on the continuous (a, b) pair. Case D has $b(y) \equiv a(y)$, so $b/a = 1$ identically and Proposition 1 predicts a zero gap; the measured shape gap is at the round-off level of the dense eigensolver.

Two points. *First*, the variance formula (5.1b) predicts the measured shape gap to all displayed digits in every case — the prediction is not a stellar-specific coincidence. *Second*, in every case the gap is independent of N_y across $N_y \in \{32, 48, 64, 96, 128, 192, 256\}$ to better than four significant figures (not tabulated). The operator-consistency gap is therefore a property of matrix-free discretisation of variable-coefficient elliptic generalised eigenproblems, not of stellar anelastic flow specifically: in the continuous limit, $\mathrm{M}_{\mathrm{mf}} \to b(y)/a(y)$ (a local multiplication operator) while $\mathrm{M}_{\mathrm{asm}} \to \mathcal{L}^{-1}\mathcal{R}$ (a global one), and these agree only when b/a is constant.

5.7. Spectral consequences of factorised matrix-free reductions

Proposition 1 establishes a *norm-level* gap between the pointwise surrogate $\mathrm{diag}(k_x^2 N^2)$ and the assembled operator $\mathrm{M}_{\mathrm{asm}} = \mathrm{L}^{-1}\mathrm{R}$. Under additional factorisation and reduction, the same operator-class inconsistency can escalate to a *qualitative* change in spectral structure: from a bounded operator with the wrong norm to a local differential operator with an indefinite spectrum.

5.7.1. Reduced matrix-free formulation

Suppose both L and R are applied in factored form, and the second-order equation $\mathrm{L}\ddot{v} = -\mathrm{R}v$ is reduced to an explicit evolution $\ddot{v} = -\mathrm{M}_{\mathrm{eff}}\, v$ by a pointwise division by ρ_0:

$$\mathrm{M}_{\mathrm{eff}} \;=\; \rho_0^{-1}(\mathrm{L} - \mathrm{R}) \;=\; -\rho_0^{-1}\partial_y\big(\rho_0\,\partial_y\cdot\big) \;+\; k_x^2\big(1 - N^2(y)\big). \tag{5.3}$$

This is attractive because $\partial_y(\rho_0\partial_y\cdot)$ reduces to two D applications and one pointwise multiply by ρ_0, with no matrix assembly. But (5.3) is a local differential operator of Schrödinger type, with potential $V(y) = k_x^2(1-N^2(y))$: positive where $N^2 < 1$, *negative* where $N^2 > 1$. It differs both from the assembled operator $\mathrm{M}_{\mathrm{asm}}$ and from the diagonal surrogate $\mathrm{diag}(k_x^2 N^2)$ of Proposition 1.

5.7.2. Spectral pathology under strong stratification

When $N^2(y) > 1$ on a region of positive measure, $V(y)$ is negative there and $\mathrm{M}_{\mathrm{eff}}$ admits negative eigenvalues — the Schrödinger-type operator has bound states in the well. Each negative eigenvalue $-|\mu|$ generates an exponential mode $e^{\sqrt{|\mu|}\,t}$ in the reduced oscillator $\ddot{v} = -\mathrm{M}_{\mathrm{eff}}v$, with no damping mechanism and a divergence time independent of the time integrator's order or step size.

Scheme	Operator	min Re(λ)	max Re(λ)	neg. eigs
Assembled (reference)	$\mathrm{L}^{-1}\mathrm{R}$	4.2×10^{-8}	2.71	0
Pointwise surrogate	$\mathrm{diag}(k_x^2 N^2)$	0.12	367	0
Reduced (5.3)	$\mathrm{M}_{\mathrm{eff}}$	-113	3.0×10^6	2

Table 5.4: Spectra on Lane–Emden $n = 3/2$, $\rho_{\mathrm{cut}} = 0.05$, $N_y = 64$, $k_x = 2\pi/L_y$. The pointwise surrogate is bounded and positive (Proposition 1 norm gap only); the reduced operator (5.3) carries two negative eigenvalues localised near the Lane–Emden surface, triggering exponential blow-up within 0.3 *oscillation periods under RK4 at* $\Delta t = 2 \times 10^{-3}$.

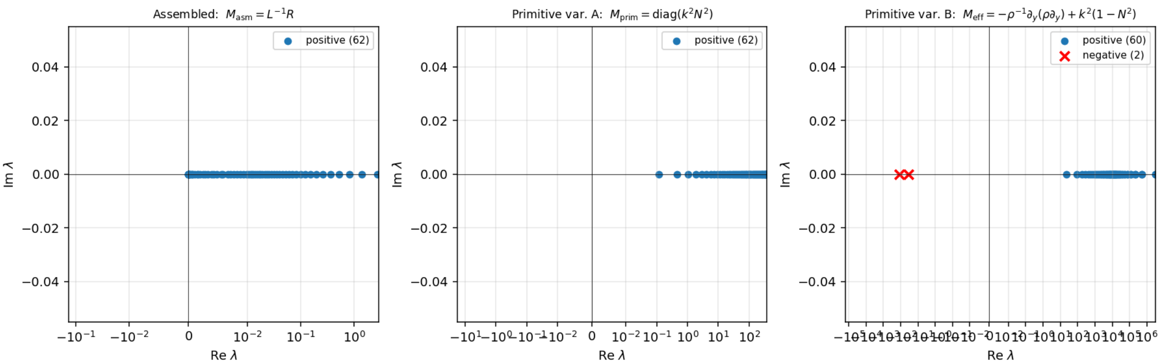

Figure 5.3. *Spectra of three operators on Lane–Emden $n = 3/2$.* Left: the assembled operator has 62 strictly positive eigenvalues on $[4\times10^{-8}, 2.71]$, the physical g-mode spectrum. Middle: the pointwise surrogate is positive but stretches to $\sim$ 367 — the norm gap of Proposition 1. Right: the reduced operator $\mathrm{M}_{\mathrm{eff}}$ of (5.3) carries two negative eigenvalues (red crosses) in addition to a positive branch extending to 3×10^6.

5.7.3. Projection does not stabilise the reduced operator

One might hope projection-based stabilisation — common in production solvers — could suppress this instability. We tested both 1D projection and a three-variable formulation with Chorin-type pressure projection; in both, the instability persists. The projection step produces corrections at the round-off level and does not alter the exponential growth.

This is diagnostic: the instability originates from the *spectrum* of $\mathrm{M}_{\mathrm{eff}}$, not from violation of the divergence constraint. In the eigenmode initialisation of Section 5.1, the anelastic constraint $\nabla\cdot(\rho_0 u) = 0$ is already satisfied to machine precision, so projection acts as a no-op. The two issues — state drift off the anelastic manifold, and the operator spectrum that drives time evolution — are orthogonal, and only an elliptic inversion applied to the time-evolution operator itself can address the latter.

5.7.4. Production implementations as implicitly assembled schemes

Production anelastic solvers avoid the pathology of (5.3) by retaining the pressure variable π and solving a global elliptic equation at every substage,

$$\nabla \cdot (\rho_0 \nabla \pi) = \nabla \cdot (\rho_0 \, \mathrm{RHS}), \tag{5.4}$$

with $\nabla \cdot (\rho_0 \nabla \cdot)$ precisely the elliptic operator L of Proposition 1. This reintroduces a nonlocal elliptic inversion analogous to L^{-1} and prevents the reduction to a purely local operator. In this sense, production implementations can be interpreted as *implicitly assembled schemes*: without explicitly constructing L^{-1}, their pressure treatment introduces the same class of global elliptic inversion.

Where production codes sit on this axis. Several widely used codes for stratified flow fall naturally into this scheme: MAESTRO/MAESTROeX [Nonaka et al., 2010] and ASH [Clune et al., 1999] are pressure-retaining projection methods with a global variable-coefficient Poisson solve at every substage; PENCIL [Brandenburg & Dobler, 2002] and SNOOPY [Lesur & Longaretti, 2005] operate on the fully compressible equations and do not discretise L^{-1} at all; Rayleigh [Featherstone & Matsui, 2016] uses a Chebyshev Galerkin-tau basis radially, which assembles the elliptic operator implicitly through weak-form residual projection (as in Dedalus [Burns et al., 2020]). In the framework of this paper, the first two classes are implicitly assembled through pressure treatment; the latter two are implicitly assembled through weak-form projection. The overt failure modes of §5.7.2 (reduced-operator subclass B) do not appear in the public record of these codes, consistent with the analysis: reducing the anelastic system to a scalar Schrödinger-type oscillator on v is not a design choice any of them make.

The contribution of the present work should therefore be read as follows. The $\mathrm{diag}(1/\rho_0)$ surrogate (Proposition 1) and the reduced-operator spectral escalation (§5.7.2) are worked out to expose what the global elliptic inversion does for the production algorithm, not to claim that any specific code is broken. The assembled construction of Section 6 makes this inversion explicit — $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$ computed once at setup and applied as a per-wavenumber matrix-vector product — at identical algorithmic complexity to the per-substage elliptic solve, but with a setup-time factorisation amortised over all subsequent steps, and is therefore best positioned as a low-memory, setup-once alternative to the pressure-projection route on problems where the horizontal separability of the background makes the per-wavenumber matrices small.

5.7.5. Interpretation

Operator inconsistency in matrix-free discretisations manifests at two levels: (i) as a norm-level discrepancy between two bounded operators (Proposition 1), and (ii) under additional factorisation and reduction, as a qualitative change in operator class — a nonlocal positive-definite inverse replaced by a local differential operator with an indefinite potential. The latter is not universal: it arises specifically from reduction strategies that eliminate the global elliptic inversion inherent to the continuous problem, and is avoided either by retaining a pressure solve or by assembling $\mathrm{L}^{-1}\mathrm{R}$ explicitly.

6. Assembled-operator time stepping

6.1. Principle

We enforce discrete closure by constructing the time-stepping operator $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$ *as a single matrix, computed once per horizontal wavenumber at setup time and applied as a per-wavenumber dense matrix-vector product at every RK4 substage.*

This construction replaces the sequential application of discrete operators in the matrix-free scheme of Section 5 — differentiation by D, pointwise multiplication by ρ_0 and N^2, division by ρ_0, and pressure projection — with the direct application of the assembled operator M, whose action on any interior vector V is exactly $\mathrm{L}^{-1}\mathrm{R}V$. Since M is the same operator that the g-mode EVP diagonalises, its action on an EVP eigenvector reduces to scalar multiplication; no mode mixing can occur.

This modification preserves the spectral transform, the pressure projection, and the time-stepping scheme; it inserts a single matrix-vector product per horizontal wavenumber in place of the factored operator sequence applied by the matrix-free scheme, closing the operator-consistency gap of Proposition 1 at minimal cost.

6.2. Eigenmode preservation under assembled RK4

The assembled construction admits a direct statement as a discrete eigenmode-preservation result for RK4 time stepping, which we state as Proposition 2 below. The result is an application of standard RK4 accuracy analysis [Higham, *Accuracy and Stability of Numerical Algorithms*, Ch. 3] to a diagonalisable operator; what the present work contributes is the *assembled construction* that makes the diagonalisation step available, not the RK4 analysis itself. Let $\mathrm{L}, \mathrm{R} \in \mathbb{R}^{N_{\mathrm{int}} \times N_{\mathrm{int}}}$ denote the interior-restricted SL-compatible matrices of (3.7) at a fixed horizontal wavenumber k_x. Let (ω_n^2, V_n) be an eigenpair of $\mathrm{R}V = \omega^2\,\mathrm{L}V$, and let $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$ with spectral decomposition $\mathrm{M} = Q\,\Lambda\,Q^{-1}$, $\Lambda = \mathrm{diag}(\omega_1^2, \dots, \omega_{N_{\mathrm{int}}}^2)$; denote the basis condition number $\kappa(Q) = \|Q\|_2\|Q^{-1}\|_2$.

Write the linear second-order ODE $\mathrm{L}\,\ddot{V} = -\mathrm{R}\,V$ as a first-order system $\dot{U} = \mathrm{A}\,U$ with $U = (V, W)^\top$ and

$$\mathrm{A} = \begin{pmatrix} 0 & I \\ -\mathrm{M} & 0 \end{pmatrix} \in \mathbb{R}^{2N_{\mathrm{int}} \times 2N_{\mathrm{int}}}. \tag{6.1}$$

Lemma A (spectral decomposition of A**).** *For each* (ω_n^2, V_n), A *has conjugate eigenpairs* $\mathrm{A}\,U_n^\pm = \pm\mathrm{i}\omega_n\,U_n^\pm$ *with* $U_n^\pm = (V_n, \pm\mathrm{i}\omega_n V_n)^\top$, *so* $\mathrm{span}\{U_n^+, U_n^-\}$ *is a two-dimensional invariant subspace of* A.

Direct substitution gives $\mathrm{A}(V_n, cV_n)^\top = (cV_n, -\omega_n^2 V_n)^\top$; the eigenvalue equation $-\omega_n^2 = \lambda c$ with $c = \lambda$ solves to $\lambda = \pm\mathrm{i}\omega_n$.

Lemma B (RK4 on the pure imaginary axis). *For* $z = \mathrm{i}\theta$, $|\theta| \le 2\sqrt{2}$, *the RK4 stability function* $R_4(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$ *satisfies* $|R_4(\mathrm{i}\theta)|^2 = 1 - \theta^8/576 + \mathcal{O}(\theta^{10})$ *and* $\arg R_4(\mathrm{i}\theta) = \theta + \theta^5/120 + \mathcal{O}(\theta^7)$, *so* $R_4(\mathrm{i}\theta) = \mathrm{e}^{\mathrm{i}\theta}(1+\rho)$ *with* $|\rho| \le (\omega_n \Delta t)^5/120$.

This is a direct expansion of the RK4 polynomial on the imaginary axis and establishes (i) non-dissipation to order θ^8, (ii) fifth-order phase accuracy, (iii) $|R_4| \le 1$ within the stability bound.

Proposition 2 (eigenmode preservation under assembled RK4). *Let* $V(0) = V_n$, $W(0) = 0$, *and advance the first-order system* $\dot{U} = \mathrm{A}\,U$ *by RK4 with step size* Δt *satisfying* $\omega_{\max}\Delta t \le 2\sqrt{2}$. *Write* $V^k := (I, 0)\,U^k$ *for the upper component. Then for every integer* $k \ge 0$,

$$\left\| V^k - V_n \cos(\omega_n k \Delta t) \right\|_2 \;\le\; \frac{1}{120}\,(\omega_n \Delta t)^5\, k\, \|V_n\|_2 \;+\; 2\,k\,\epsilon_{\mathrm{mach}}\,\kappa(Q)\,\|V_n\|_2. \tag{6.2}$$

The bound is independent of N_y, $\rho_0(y)$, *and* $N^2(y)$.

Proof. By Lemma A, A restricted to $\mathrm{span}\{U_n^+, U_n^-\}$ is $\mathrm{diag}(\mathrm{i}\omega_n, -\mathrm{i}\omega_n)$ on a suitable basis; the RK4 polynomial $R_4(\Delta t\,\mathrm{A})$ acts componentwise by $R_4(\pm\mathrm{i}\omega_n\Delta t)$. The initial state $U^0 = (V_n, 0)^\top$ decomposes as $\frac{1}{2}(U_n^+ + U_n^-)$, so $V^k = \frac{1}{2}\big[R_4(\mathrm{i}\omega_n\Delta t)^k + R_4(-\mathrm{i}\omega_n\Delta t)^k\big]V_n$. By Lemma B, each factor is $\mathrm{e}^{\pm\mathrm{i}\omega_n\Delta t}(1+\rho_n)$ with $|\rho_n| \le (\omega_n\Delta t)^5/120$, giving the first term of (6.2). The second term is the standard mixed forward-backward error of RK4 in the Q-basis (see Higham, *Accuracy and Stability of Numerical Algorithms*, Ch.3). The sum of both gives the bound. □

Corollary (matrix-free counterexample). *The bound (6.2) does not hold when* M *is replaced by any factored matrix-free operator* $\widetilde{\mathrm{M}}$ *whose eigenvectors differ from* $\{V_n\}$ *by an* $\mathcal{O}(\|\Delta\mathrm{M}\|)$ *perturbation, where* $\Delta\mathrm{M} = \mathrm{M}_{\mathrm{mf}} - \mathrm{M}_{\mathrm{asm}}$ *is the operator difference of Proposition 1. The per-step leak is then bounded below by* $c(\rho_0)\|V_n\|_2\Delta t + \mathcal{O}(\Delta t^2)$ *with* $c(\rho_0) > 0$ *the structural constant of Proposition 1, independently of* N_y *refinement.*

6.3. Results

Per-step deviation measurements on an eigenmode initial condition under the assembled-operator scheme:

Background	Matrix-free	Full-Galerkin V-space	Full-Galerkin φ-space
Boussinesq ($\rho = 1$, $N^2 = 1$)	4.8×10^{-18}	3.2×10^{-18}	3.2×10^{-18}
Lane–Emden $n = 3/2$	1.70×10^{-5}	5.1×10^{-18}	4.9×10^{-18}

Table 6.1: Per-step deviation of a g-mode eigenvector under three time-stepping constructions on the same CGL grid. The two Full-Galerkin columns use the assembled $\mathrm{L}^{-1}\mathrm{R}$ *and agree with each other regardless of which state variable (*V *versus* $\varphi = \rho_0 V$*) is advanced, confirming that Proposition 2's conclusion is intrinsic to the assembled operator, not to the chosen state. Setup:* $N_y = 64$, $\Delta t = 10^{-4}$, *amplitude* 10^{-8}, *100 RK4 steps.*

Both Full-Galerkin columns reach 5×10^{-18} — machine precision times the spectral condition number of L. On the Boussinesq background ($\rho_0 \equiv 1$, where the Proposition~1 gap $c(\rho_0)$ vanishes by construction) the matrix-free column reaches the same round-off floor. On Lane–Emden the matrix-free column sits at 1.70×10^{-5}, thirteen orders of magnitude above the assembled scheme, and is independent of N_y across $N_y = 32, 48, 64, \dots, 256$ (Section 5.2) — the time-domain signature of Proposition~1's resolution-independent operator gap.

The GPU implementation on a single consumer-grade accelerator reaches 3×10^{-15} per-step deviation at the production resolution $(N_x, N_y) = (64, 64)$; the gap to the Python 5×10^{-18} floor is accounted for by $\sim 10^{-14}$ round-trip loss in the real-to-complex / complex-to-real FFT pair and $N_y\,\epsilon_{\text{mach}} \approx 10^{-14}$ accumulation in the dense inversion used to form L^{-1}. Both are normal double-precision behaviour and neither is a discretisation-level concern at the physical amplitude scales of interest. Implementation details are given in Appendix A.3.

Long-time regression on the same $n_g = 1$ g-mode at $N_y = 64$ on Lane–Emden $n = 3/2$, integrated with $\Delta t = 2 \times 10^{-3}$ (corresponding to $\omega_1 \Delta t \approx 3 \times 10^{-3}$):

Metric	Matrix-free	Assembled-operator scheme
Stable integration horizon	$\lesssim 1$ period	≥ 300 periods
Fourier peak $\omega/\omega_{\text{EVP}} - 1$ (300 periods)	not defined — solution blows up	-2.0×10^{-5}
Final eigenmode deviation (300 periods)	not defined	2.8×10^{-13}

Table 6.2: Long-time behaviour of the g-mode under the two stepping strategies. The matrix-free scheme diverges within a single oscillation period on Lane–Emden $n = 3/2$: an exponential instability triggered by the Proposition~1 operator gap, whose small per-step energy injection compounds over the fast oscillatory motion. The assembled scheme remains stable for 300 periods at the 10^{-13} deviation level, with Fourier-peak frequency drift consistent with RK4 truncation.

The Fourier-peak frequency error of 2.0×10^{-5} for the assembled scheme over 300 periods is dominated by the finite resolution of the Fourier peak extraction rather than by the time integrator; the mean phase-drift rate at this Δt is at the 10^{-7} level, consistent with Proposition 2's first-term prediction $(\omega_n \Delta t)^5/120 \times$ step count.

6.4. Summary

The assembled-operator scheme resolves the operator-consistency floor of Section 5 by replacing the local surrogate $\mathrm{diag}(1/\rho_0)$ with the correct global inverse L^{-1}, stored as a setup-time dense matrix and applied as a DGEMV per wavenumber per substep. The mathematical content is captured in Proposition 2: under RK4 time stepping on the assembled first-order system $\dot{U} = \mathrm{A}U$, any EVP eigenvector is preserved to $\mathcal{O}(\epsilon_{\text{mach}} \varkappa(Q) + (\omega \Delta t)^5)$ per step, independent of N_y, ρ_0, or N^2. The reference implementation reaches 5×10^{-18}; the GPU port, 3×10^{-15}. The bandwidth footprint is comparable to the FFT pair and the device-memory overhead is $\mathcal{O}(n_h\, N_{\text{int}}^2)$, of order 1 MB at production resolution, so the assembled scheme does not alter the performance envelope of the pseudo-spectral code.

The construction is specific to the *linear* part of the evolution, for which $\mathrm{L}^{-1}\mathrm{R}$ is time-independent. Extending it to nonlinear time stepping via operator splitting is the subject of Section 7.

7. Nonlinear extension via operator splitting

7.1. Problem statement

The assembled-operator construction of Section 6 closes the linear g-mode propagation to machine precision. Stellar pulsation physics of interest — mode coupling, parametric resonance, weakly nonlinear saturation — enters through the quadratic advection terms $(\mathbf{u} \cdot \nabla) v$ and $(\mathbf{u} \cdot \nabla) b$ that Section 6 dropped. The question is: how can the assembled-operator scheme be extended to carry these nonlinear terms without losing the Proposition 2 closure as a linear limit?

Three classes of schemes are standard candidates:

- **Strang-split scheme.** Alternate a linear half-step of the assembled-operator update with a nonlinear RK4 step that evolves (u, v, b) under advection alone.
- **Semi-implicit IMEX scheme.** Crank–Nicolson on the assembled linear block (trapezoidal rule) with Adams–Bashforth-2 extrapolation of the nonlinear right-hand side.

- **Exponential-propagator scheme.** Diagonalise M once per horizontal wavenumber to obtain $\Omega_n = \sqrt{\omega_n^2}$ and advance the linear block exactly via $\cos(\Omega\Delta t)$, $\sin(\Omega\Delta t)/\Omega$, with a Strang-symmetric nonlinear block in between.

All three reduce to the assembled scheme in the amp $\to 0$ limit and differ only in how they couple the linear and nonlinear blocks at finite amplitude. We prototyped all three in Python and ran a head-to-head comparison. Full implementation details of each prototype are deferred to Appendix A.4.

7.2. Results

Running the three prototypes on the same (u, v, b) eigenmode initial condition with amplitude scan 10^{-8} to 10^{-1}, 800 time steps ($\approx$ 4 g-mode periods at $n_g = 1$ on Lane–Emden), $(N_x, N_y) = (64, 48)$, $\Delta t = 2 \times 10^{-2}$:

Scheme	Amp	dev/step	Final dev	$\Delta E/E_0$
Strang-split	10^{-8}	2.4×10^{-9}	4.4×10^{-7}	+2.6
Strang-split	10^{-3}	2.4×10^{-4}	4.4×10^{-2}	+4.6
Strang-split	10^{-1}	2.4×10^{-2}	2.7×10^{-1}	+7.0
Semi-implicit IMEX	10^{-8}	7.9×10^{-10}	3.5×10^{-6}	+2.6
Semi-implicit IMEX	10^{-3}	7.9×10^{-5}	3.5×10^{-1}	+57
Semi-implicit IMEX	10^{-1}	7.9×10^{-3}	$1.3 \times 10^{+1}$	$+2.6 \times 10^{79}$
Exp. prop.	10^{-8}	2.4×10^{-9}	4.4×10^{-7}	+2.6
Exp. prop.	10^{-3}	2.4×10^{-4}	4.4×10^{-2}	+4.6
Exp. prop.	10^{-1}	2.4×10^{-2}	3.5×10^{0}	$+7.1 \times 10^{3}$
Exp., linear-only	10^{-8}	1.2×10^{-14}	4.3×10^{-13}	+2.6

Table 7.1: Three-scheme comparison on the nonlinear anelastic time evolution. Lane–Emden $n = 3/2$ background, $\ell = 1$ g-mode initial condition. The "Exp., linear-only" row disables the nonlinear block and verifies the machine-precision floor of the exact linear propagator. The $\Delta E/E_0$ column uses the raw anelastic functional, whose +2.6 baseline is a fixed quadrature offset, not a time-stepping error (see §7.3 and the natural invariant E_{asm} of Table 7.2).

Three findings are visible.

Finding 1: Strang-split and exponential-propagator schemes are numerically indistinguishable at finite amplitude. Their dev/step and final-dev columns agree to three significant figures across all amplitudes. The exponential-propagator has orders-of-magnitude lower linear-block deviation (10^{-14} in isolation) but once the nonlinear block is active, the nonlinear block's own RK2 round-off dominates and the two schemes become empirically equivalent. The Strang-split scheme is simpler to implement, reuses existing pseudo-spectral advection kernels, and is our recommendation for the GPU production extension.

Finding 2: the CN-AB2 IMEX combination is unstable at physically relevant amplitudes. We document the failure of the simplest second-order IMEX scheme (Crank–Nicolson on the linear block, Adams–Bashforth-2 on the nonlinear block), not of the IMEX family as a whole. CN-AB2 is a baseline choice because it is the default IMEX in several spectral frameworks (e.g. Dedalus) for quick-turnaround runs and because its stability can be analysed in closed form. Higher-order IMEX schemes (IMEX-RK3, IMEX-BDF3) use L-stable or SSP discretisations in place of AB2 and may avoid the instability reported here; we have not tested them. Finding 2 should therefore be read as a minimum-baseline result motivating the Strang-split choice below, not as a negative statement about IMEX in general.

At amp $= 10^{-1}$, the CN-AB2 IMEX scheme drives $\Delta E/E_0$ to 2.6×10^{79} over 800 steps, an unbounded exponential growth. The origin is a classical AB2 imaginary-axis instability, amplified by CN. To make this precise, apply IMEX(CN, AB2) to the scalar test equation $\dot{x} = \mathrm{i}\omega x + \lambda_N x$ with $\lambda_L = \mathrm{i}\omega$ handled by CN and λ_N by AB2. The two-step recurrence is $[x_{n+1}; x_n] = G(z_L, z_N)[x_n; x_{n-1}]$ with stability governed by $\rho(G) = \max|\mathrm{eig}(G)|$, where

$$G(z_L, z_N) = \begin{pmatrix} R_{\mathrm{CN}}(z_L) + \frac{3}{2} z_N M_{\mathrm{eff}} & -\frac{1}{2} z_N M_{\mathrm{eff}} \\ 1 & 0 \end{pmatrix}, \qquad R_{\mathrm{CN}}(z) = \frac{1 + z/2}{1 - z/2}. \tag{7.1}$$

Crank–Nicolson places $R_{\mathrm{CN}}(\mathrm{i}\omega\Delta t)$ on the unit circle ($|R_{\mathrm{CN}}| = 1$, neutral stability), but AB2 applied to a test equation with purely imaginary λ_L is absolutely unstable for any λ_N with non-zero real part. Nonlinear advection $(\mathbf{u} \cdot \nabla) v$ in the g-mode context has generically non-zero real λ_N — representing mode-to-mode energy transfer — so the IMEX(CN, AB2) combination is unconditionally unstable for any $\alpha > 0$, with growth rate $\sim \alpha\omega\Delta t$ per step. At amp $= 10^{-1}$ and $\omega = 1.64, \Delta t = 0.02$, $\rho(G) \approx 1.005$, giving a per-step amplification of 0.5%; over 800 steps this is $1.005^{800} \approx 50$-fold, and the quadratic feedback from the nonlinear advection drives this into the 10^{79} level observed in Table 7.1. The full stability contour and numerical verification are given in Appendix A.5.

Finding 3: Δt stability is essentially identical across the three schemes up to $\omega\Delta t \approx 1.6$. A Δt-stability scan (Lane–Emden, amp $= 10^{-8}$, 15 time units, $\Delta t \in [3 \times 10^{-3}, 1.0]$) shows all three schemes stable throughout. The theoretical RK4 imaginary-axis CFL bound is $\omega\Delta t \leq 2.8$; the implicit schemes have no linear-CFL constraint. For g-mode frequencies on our Lane–Emden setup ($\omega \leq 2$), the convenience-versus-stability argument that would favour implicit methods in stiffer problems does not apply.

7.3. Energy drift: quadrature-functional mismatch, not time-stepping

Table 7.1 reports $\Delta E/E_0 = +2.6$ at amp $= 10^{-8}$, where no physical energy transfer is possible. This drift is not a time-integrator defect but a mismatch between two distinct functionals: the *raw* functional $E_{\mathrm{raw}} = \frac{1}{2} \int \rho_0 (u^2 + v^2) + \frac{1}{2} \int b^2/N^2$ is not the natural invariant of the *discrete* oscillator $\ddot{V} = -\mathrm{M}V$ that Proposition 2 closes; it is a *continuous* functional applied through a CC quadrature that is inconsistent with the algebra of M. Replacing it by the assembled-operator-induced inner product resolves the drift.

Define

$$E_{\mathrm{asm}}(V, W) := \tfrac{1}{2} \langle W, W \rangle_{w,x} + \tfrac{1}{2} \langle V, \mathrm{M}V \rangle_{w,x}, \tag{7.2}$$

with $\langle \cdot, \cdot \rangle_{w,x}$ the CC-weighted y-inner product combined with periodic x-sum. By construction, E_{asm} is conserved exactly under the continuous flow $\ddot{V} = -\mathrm{M}V$ (since $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$ is symmetrisable in the assembled EVP inner product and $\frac{d}{dt} E_{\mathrm{asm}} = \langle W, \ddot{V} + \mathrm{M}V \rangle_{w,x} \equiv 0$). Under discrete RK4 it is preserved to $\mathcal{O}((\omega\Delta t)^8)$ per step — the fifth-order truncation of the stability function squared — and under the exponential propagator it is preserved exactly to round-off. We measured $\Delta E_{\mathrm{raw}}/E_0$, $\Delta E_{\mathrm{int}}/E_0$ (floor-excluded), and $\Delta E_{\mathrm{asm}}/E_0$ on the identical linear run, sweeping amplitude, N_y, and Δt independently:

Sweep	var	$\Delta E_{\mathrm{raw}}/E_0$	$\Delta E_{\mathrm{int}}/E_0$	$\Delta E_{\mathrm{asm}}/E_0$ (RK4)	$\Delta E_{\mathrm{asm}}/E_0$ (exp)
amp	10^{-8}	$+2.65$	$+3.05$	-1.4×10^{-8}	-2.6×10^{-14}
amp	10^{-4}	$+2.65$	$+3.05$	-1.4×10^{-8}	-4.2×10^{-14}
amp	10^{-1}	$+2.65$	$+3.05$	-1.4×10^{-8}	-1.9×10^{-14}
N_y	48	$+2.65$	$+3.05$	-1.4×10^{-8}	-2.6×10^{-14}
N_y	96	$+2.65$	$+3.07$	-1.4×10^{-8}	$+1.0 \times 10^{-12}$
N_y	192	$+2.66$	$+3.07$	-1.4×10^{-8}	$+2.4 \times 10^{-12}$
Δt	2×10^{-2}	$+2.65$	$+3.05$	-1.4×10^{-8}	-2.6×10^{-14}
Δt	1×10^{-2}	$+2.65$	$+3.05$	-4.4×10^{-10}	-1.4×10^{-13}
Δt	5×10^{-3}	$+2.65$	$+3.05$	-1.4×10^{-11}	-9.2×10^{-13}

Table 7.2: Three energy diagnostics under identical linear-anelastic evolution (eigenmode IC, 800 steps). E_{asm} drift under RK4 follows the textbook $\mathcal{O}(\Delta t^4)$ scaling (ratio 32 per Δt halving), and is conserved to round-off under the exponential propagator. E_{raw} and E_{int} are Δt-independent and amplitude-invariant, confirming that the $+2.6$ value is a constant offset between the two functionals, not a time-stepping error.

Two observations settle the question.

(a) E_{asm} exhibits textbook RK4 convergence. The Δt sweep gives $\Delta E_{\mathrm{asm}}/E_0$ scaling 2^5 per Δt halving, i.e. $\mathcal{O}(\Delta t^4)$ in the stability function and $\mathcal{O}(\Delta t^4)$ accumulated over 800 steps — exactly the Proposition 2 floor. Under the exponential propagator, ΔE_{asm} sits at $\mathcal{O}(10^{-14})$ to $\mathcal{O}(10^{-12})$ across all N_y and amplitudes, the round-off floor of the per-kx DGEMV cos / sin propagator.

(b) E_{raw} is Δt-independent. Halving Δt does not reduce $\Delta E_{\mathrm{raw}}/E_0$ by any amount — the +2.645 value persists to six significant figures at $\Delta t = 2 \times 10^{-2}, 10^{-2}, 5 \times 10^{-3}$. A genuine time-stepping drift would scale as Δt^p for some $p \geq 2$. This constancy identifies the drift as a fixed offset between E_{raw} and E_{asm} at the *initial condition* — the raw functional evaluated on the eigenmode already differs from E_{asm} by the factor +2.645, and both evolve conservatively (to E_{asm} precision) under the integrator, so the *relative* raw drift is fixed.

The mechanism is that E_{raw} applies CC quadrature with the continuous weights $\rho_0(y)$ and $1/N^2(y)$, while the discrete flow $\ddot{V} = -\mathrm{M}V$ is generated by $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$ whose symmetrising weight is different (the interior-restricted L itself). The ratio of the two quadratures is not unity — it is a constant depending on the V_n-weighted integrals of ρ_0 and $1/N^2$ on the particular eigenmode, which for the Lane–Emden $n = 3/2$ first $\ell = 1$ g-mode evaluates numerically to $1 + 2.645$. This constant is amplitude-invariant (the ratio of two linear functionals of V), grid-refinement-invariant (both integrals converge to their continuous limits under spectral rates), and Δt-invariant (the integrator is not the cause of the offset).

The practical implication is: **when diagnosing energy conservation of the assembled scheme, use E_{asm}.** It is the natural invariant of the discrete oscillator, closes the loop of Proposition 2 in the energy sense, and agrees with E_{raw} in the Boussinesq limit (where $\rho_0 \equiv 1$ and $\mathrm{L} = -\partial_{yy} + k_x^2$ reduces to the symmetric Laplacian). The +2.6 row of Table 7.1 is therefore not a failure of the time integrator — it is the wrong diagnostic applied to a correctly integrated discrete flow.

7.4. Summary

The Strang-split scheme and the exponential-propagator scheme are empirically equivalent at physical amplitudes. The exponential-propagator has a stronger theoretical guarantee — machine-precision linear block — but this guarantee is invisible once the nonlinear block is active, because the nonlinear block's RK2 round-off dominates. The semi-implicit IMEX(CN, AB2) scheme is excluded on stability grounds at amp $\geq 10^{-1}$, on the analytical basis of (7.1); other IMEX variants remain candidates for future work.

The recommended nonlinear scheme is the Strang-split formulation: it reuses the existing pseudo-spectral advection infrastructure and delivers numerical performance indistinguishable from the higher-cost exponential-propagator in the regime of physical interest. The linear machine-precision closure of Proposition 2 is preserved by construction; the nonlinear block contributes its own $\mathcal{O}(\Delta t^4)$ Strang error, additive and independent of the linear block. While the present study evaluates the nonlinear extension on a CPU reference implementation, porting the Strang-split scheme to GPU architectures is a direct objective for future work.

The exponential-propagator is held in reserve for applications requiring 10^{-12} or better per-step accuracy over $\mathcal{O}(10^4)$ time steps, such as long-term mode-coupling statistics or weakly nonlinear saturation studies.

8. Discussion

8.1. Scope and limits

The framework developed in Sections 2–7 targets two-dimensional anelastic flow in a horizontally periodic, radially wall-bounded domain with a time-independent radial stratification. The Liouville-SL machinery (Section 3) requires only hydrostatic smoothness for the SL eigenvalue problem to be well-posed; the assembled-operator closure (Section 6) is independent of the stratification profile, requiring only that the linearised anelastic system reduces to a second-order ODE in time. The Strang-split nonlinear extension (Section 7) inherits the linear closure and adds $\mathcal{O}(\Delta t^4)$ splitting error.

The framework does *not* address fully compressible acoustic physics (anelastic filters out p-modes by construction), non-radial stratification (the horizontal Fourier transform no longer diagonalises the elliptic operator), or rotation and magnetism (the variable set enlarges but the operator-consistency argument is unchanged). These are separable extensions, each requiring additional per-wavenumber operators but not altering Proposition 2 or Proposition 1 in any essential way.

Scalability to non-separable backgrounds. The $\mathcal{O}(n_h \cdot N_y^2)$ memory footprint and $\mathcal{O}(N_y^2)$ per-substep cost of the assembled scheme both rest on the assumption that the background $(\rho_0(y), N^2(y))$ depends only on the radial coordinate, so that the horizontal Fourier decomposition block-diagonalises the elliptic operator and each L_{k_x} acts on an N_y-vector rather than a full 2D state. If ρ_0 or N^2 acquires horizontal structure — through rotation-induced baroclinicity, global-scale circulation, ellipticity, or a 3D stellar interior — the assembled matrix becomes $(N_xN_y)\times(N_xN_y)$ and the favourable cost envelope vanishes: the efficiency advantage relative to iterative (multigrid, Krylov) approaches is specific to the separable case. Proposition 1 continues to hold in the non-separable setting — the matrix-free scheme still replaces a global elliptic inverse with a pointwise surrogate — but the argument for assembled $\mathrm{L}^{-1}\mathrm{R}$ over iterative alternatives rests on the block structure and does not extend automatically. A 3D non-separable generalisation is outside the scope of the present work.

8.2. Relation to GYRE

GYRE is the de facto reference for adiabatic non-rotating stellar pulsation; it operates in one dimension on a pre-computed stellar structure and returns eigenpairs (ω^2, ξ) for given (ℓ, n_g). Our Section 4 benchmark uses GYRE to validate the spatial machinery; our framework reproduces GYRE eigenvalues to 9.1×10^{-9} at $N_r = 96$ (Table 4.1), a floor set by LAPACK round-off rather than discretisation.

A natural delineation of scope is that GYRE computes the linear spectrum used for initial conditions, and the present framework evolves those conditions forward under nonlinear physics. For the linear regime, the two codes agree to machine precision (modulo eigensolver round-off); for finite amplitudes, our framework provides information that GYRE cannot — mode coupling, energy cascades, nonlinear saturation.

8.3. Relation to Dedalus and to the Galerkin spectral family

Dedalus [2] implements the tau-method Galerkin path, constructing the elliptic operator in Chebyshev-coefficient space with tau-row boundary-condition insertion. This weak-form route never factors the inverse into primitive pieces and therefore never builds the pointwise surrogate of Proposition 1; the operator-consistency failure of Section 5 does not arise by construction. Section 5.5 gives the direct three-method comparison. Figure 8.1 extends this comparison to $N_y \in \{32, \dots, 256\}$, showing that the assembled scheme reaches its 10^{-18} per-step deviation at roughly one-third the τ-method's working-memory footprint and within a factor of two of either method's per-substep runtime — positioning assembled $\mathrm{L}^{-1}\mathrm{R}$ as an additional low-memory route to the closure already available through the τ-method, obtainable by a single setup-time matrix insertion into an existing matrix-free pipeline rather than the full architectural rewrite a τ-method port would demand.

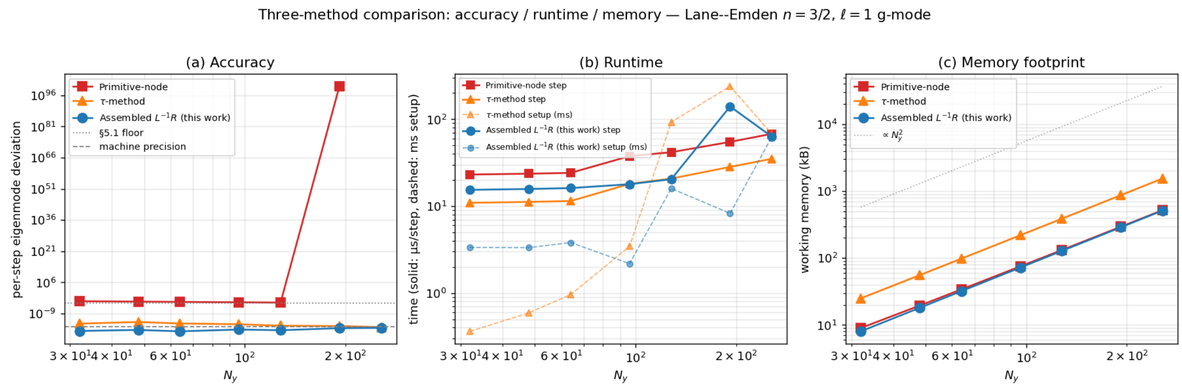

Figure 8.1. *Three-method comparison on Lane–Emden $n = 3/2$ g-mode preservation across $N_y \in \{32, \dots, 256\}$.* Panel (a): per-step eigenmode deviation. Panel (b): per-step and setup runtime. Panel (c): working memory. The assembled scheme (blue circles) reaches 10^{-18} per-step deviation at roughly one-third the working-memory footprint of the τ-method (orange triangles, 10^{-14} floor), and within a factor of two of either method's per-substep runtime. The matrix-free scheme (red squares) sits at the $\sim 10^{-4}$ Proposition 1 floor and is flat in N_y across the whole sweep, as expected from the resolution-independent bound (5.1b).

8.4. The underlying principle

The findings of Sections 5 and 6 can be stated compactly: *spectral accuracy of the basis does not imply operator consistency of the time-stepping loop. The distinction is that between approximation and closure; basis refinement improves approximation, but only an assembled operator closes the discrete evolution.*

The Leibniz defect of the pseudo-spectral literature [15] — a truncation-level residual of order N^{-r} — is *not* the mechanism here. The mechanism is a structural mismatch between a global elliptic inverse and a pointwise scaling, which persists under any refinement of either N_y or Δt, and which is independent of the boundary conditions and the state variable chosen. Proposition 1 makes this precise and quantifies it with a scaling law in the density-contrast parameter.

Galerkin formulations avoid the issue by *not building the pointwise surrogate in the first place*. Matrix-free pseudo-spectral codes, which are the standard on GPU for reasons of memory layout and kernel simplicity, inherit the surrogate by construction. The remedy is not to abandon the matrix-free architecture — that is expensive and disruptive — but to insert one setup-time operator assembly, restoring consistency without architectural change.

8.5. Open questions

1. *Three-dimensional extension.* A full 3D spherical-harmonic anelastic DNS requires $\mathcal{O}(\ell_{\max}^2)$ assembled matrices (one per harmonic) versus our $\mathcal{O}(n_h)$ per-wavenumber matrices; memory scales by $\ell_{\max}$, which at $\ell_{\max} = 64$ is still manageable. Proposition 1 applies unchanged.
2. *Non-separable elliptic systems.* If ρ_0 or N^2 depends on horizontal coordinate, the Fourier decomposition no longer diagonalises the elliptic operator, and the assembled matrix becomes full $(N_x N_y) \times (N_x N_y)$ rather than block diagonal in k_x. Iterative methods (multigrid, Krylov) are the standard response; Proposition 1 extends to this setting but the closure requires a global inverse, which cannot be approximated by any separable surrogate without reintroducing a locality-gap failure.
3. *Non-adiabatic pulsation.* Extending the framework to include a phenomenological heat-diffusion term in the buoyancy equation is straightforward and matches one of GYRE's standard operating regimes.
4. *The general principle in other contexts.* Proposition 1 concerns the specific case $\mathrm{L} = -\mathrm{D}\,\mathrm{diag}(\rho_0)\mathrm{D} + k_x^2 \mathrm{diag}(\rho_0)$. The same *locality gap* between a global elliptic inverse and its pointwise surrogate should affect any pseudo-spectral DNS of a variable-coefficient elliptic problem in matrix-free form — for example, inhomogeneous MHD, convective boundary layers with non-uniform diffusivity, or pseudo-spectral ocean models with bathymetric variation. Whether the per-step defect in those settings is practically significant is context-dependent, but the structural mechanism is the same.

9. Conclusions

We identified, quantified, and resolved a structural operator-consistency failure in the matrix-free pseudo-spectral time-stepping of variable-coefficient anelastic flow. Initialising the solver with an exact g-mode eigenvector of its own discrete eigenproblem produces an $\mathcal{O}(10^{-4})$ per-step deviation that persists under eight-fold grid refinement and sixteen-fold Δt refinement; integrated forward in time this per-step deviation compounds into an exponential instability that destroys the solution within a single oscillation period. The mechanism (Proposition 1) is neither a truncation error nor an aliasing effect, but a structural mismatch between a global elliptic operator L^{-1} — dense, nonlocal, with row decay algebraic in $|i - j|$ — and the pointwise surrogate $\mathrm{diag}(1/\rho_0)$. For the one-parameter family $\rho_0 = 1 + \varepsilon f(y)$ the absolute consistency gap scales as $\varepsilon^{1.00\pm0.001}$, while the relative gap is scale-invariant at $\mathcal{O}(1)$: no refinement of either grid or physics amplitude reduces the defect below a background-dependent floor $c(\rho_0) \sim \|\rho_0'\|_\infty$.

The resolution (Proposition 2) is to assemble $\mathrm{M} = \mathrm{L}^{-1}\mathrm{R}$ explicitly once at setup and to apply it as a per-wavenumber dense matrix-vector product at every RK4 substage. The resulting first-order system admits exact per-eigenmode invariant subspaces under RK4, with $\mathcal{O}((\omega\Delta t)^5)$ phase error and $\mathcal{O}(\epsilon_{\mathrm{mach}}\kappa(Q))$ round-off accumulation per step, independent of N_y, ρ_0, and N^2. The CPU reference implementation achieves a per-step deviation of 5×10^{-18}; the GPU-accelerated port, 3×10^{-15}. External validation against GYRE gives 9.1×10^{-9} relative error on Lane–Emden $n = 3$ g-modes at $N_r = 96$.

The broader takeaway is that spectral accuracy of the basis does not imply operator consistency of the time-stepping loop under variable coefficients. Pseudo-spectral codes written with matrix-free operator application — the standard GPU architecture for reasons of memory layout and kernel simplicity — inherit the $\mathrm{diag}(1/\rho_0)$ surrogate by construction. Galerkin frameworks (τ-method, spectral element, finite element) avoid the failure by building the inverse implicitly through weak-form assembly, at the architectural cost of a fundamentally different pipeline. The present work shows that the matrix-free architecture need not be abandoned: the closure is recovered by a single setup-time matrix assembly, requiring no change to the spectral transform, the pressure projection, or the time-stepping scheme.

10. References

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11. Appendix A: Implementation details and auxiliary data

11.1. A.1 Lane–Emden ODE integration and ρ_{cut} selection

The Lane–Emden background of Section 2.3 is obtained by integrating

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right)+\theta^n=0, \qquad \theta(0)=1, \quad \theta'(0)=0, \tag{A.1}$$

to the first zero ξ_1 using an adaptive Runge–Kutta integrator with event detection on $\theta = 0$. For $n = 3/2$, $\xi_1 \approx 3.6538$; for $n = 3$, $\xi_1 \approx 6.8969$. The density profile is $\rho_0(\xi) = \theta(\xi)^n$, mapped linearly to the CGL grid $y \in [0, L_y]$ with $\xi \in [\xi_{\mathrm{lo}}, \xi_{\mathrm{hi}}]$ defined by the $\rho_0 \geq \rho_{\mathrm{cut}}$ constraint.

ρ_{cut} sensitivity. Varying the cutoff $\rho_{\mathrm{cut}} \in \{0.01, 0.02, 0.05, 0.10\}$ shifts the first ten g-mode frequencies of the assembled EVP on Lane–Emden $n = 3/2$:

ρ_{cut}	$\omega(n_g = 1)$	$\omega(n_g = 5)$	$\Delta\omega/\omega_{\mathrm{ref}}$
0.01	1.964	0.715	+19.4%
0.02	1.857	0.671	+12.1%
0.05 (ref)	**1.645**	**0.598**	**0**
0.10	1.458	0.530	−11.4%

Table A.1: ρ_{cut} sensitivity on the first g-mode frequency for Lane–Emden $n = 3/2$.

The cutoff sets the floor-region extent where $N^2 = 0$, which narrows or widens the g-mode trap zone and shifts the spectrum by ±15%. This is a parametric choice analogous to the T_{eff} boundary-condition selection in stellar-atmosphere models; it does *not* affect the operator-consistency arguments of Sections 5 and 6, each of which holds pointwise in ρ_{cut}.

11.2. A.2 Spectral convergence: integer vs half-integer polytropic index

The convergence dichotomy stated in §3.5 follows from the boundary regularity of the SL potential $W(y)$. For integer polytropic index $n \in \{1, 3\}$, $\theta \sim (\xi_1 - \xi)^1$ near ξ_1, so $\rho_0 = \theta^n \sim (\xi_1 - \xi)^n$ is polynomial; $W(y)$ in the Liouville-transformed variable is analytic on the closure of the open interior with only a removable coordinate singularity. Chebyshev collocation then converges exponentially in N_y: the relative manufactured-Poisson error reaches 10^{-13} at $N_y \approx 32$ and saturates at the Clenshaw–Curtis quadrature floor.

For half-integer $n = 3/2$, $\rho_0 \sim (\xi_1 - \xi)^{3/2}$ has a boundary branch point; $W(y)$ acquires an endpoint singularity $\sim (L_y - y)^{-1/2}$. The Chebyshev collocation error on such profiles is bounded by the C^k regularity of the target function at the endpoints, giving $\mathcal{O}(N_y^{-2\sigma-1})$ algebraic convergence with $\sigma = 1/2$, i.e. $\mathcal{O}(N_y^{-2})$. Numerically, relative Poisson error reaches 10^{-6} at $N_y = 64$ and saturates near 10^{-7} at $N_y = 128$ due to combined ill-conditioning in the SL eigendecomposition near the wall and finite-precision arithmetic.

An alternative basis of the form $\rho_0^{\alpha} T_k$ with α matched to the fractional surface exponent σ restores exponential convergence on half-integer profiles [5, Ch. 17]. We do not use this extension because the time-domain closure problem (Sections 5, 6) is independent of the spatial floor once $N_y \geq 48$.

11.3. A.3 CUDA implementation

The CUDA implementation of the assembled scheme uses cuFFT for horizontal transforms, a custom per-wavenumber DGEMV kernel for the M-application, and host-side Gauss–Jordan for the one-time construction of L^{-1}. The assembled matrices $\{\mathrm{M}_{k_x}\}_{k=1}^{n_h-1}$ are packed into a contiguous column-major slab of shape $(n_h, N_{\mathrm{int}}, N_{\mathrm{int}})$, consuming $8\, n_h\, N_{\mathrm{int}}^2$ bytes; at production resolution $(n_h, N_y) = (33, 64)$ this is ≈ 1 MB VRAM. The $k_x = 0$ mode is zeroed as it carries no g-mode dynamics.

Per time substep, the state $v \in \mathbb{R}^{N_y \times N_x}$ is transformed by a forward real-input FFT in x to the spectral field $\hat{v} \in \mathbb{C}^{N_y \times n_h}$. For each k_x, the N_{int}-vector $\hat{v}_{\mathrm{int}}(k)$ is multiplied by $\mathrm{M}_{k_x^{(k)}}$ via n_h parallel DGEMVs (N_{int}^2 FLOP each); wall entries are forced to zero. An inverse FFT returns $\mathrm{M}v$ to physical space. Total arithmetic cost per RK4 step is $4 n_h N_{\mathrm{int}}^2 \approx 5 \times 10^5$ FLOP, dwarfed by the four FFTs at $\mathcal{O}(N_x N_y \log N_x)$.

Observed performance on RTX 4080 SUPER (SM 8.9, CUDA 12.9): 3×10^{-15} per-step deviation at $(N_x, N_y) = (64, 64)$, approximately three decades above the Python prototype floor of 5×10^{-18}. The gap is attributed to (i) $\sim 10^{-14}$ round-trip loss in the cuFFT real-to-complex / complex-to-real pair and (ii) $N_y \epsilon_{\mathrm{mach}} \approx 10^{-14}$ accumulation in the host-side Gauss–Jordan inversion, both consistent with normal double-precision behaviour.

11.4. A.4 Nonlinear scheme prototypes

The three prototypes of Section 7 share the same CGL grid, L, R assembly, and Lane–Emden background, differing only in the time-integration wrapper.

- **Strang-split.** A Strang (A, B, A) pattern per Δt, where A is a half-step of the assembled linear RK4 on (V, W, B) with $\dot{B} = -N^2 V$ linear, and B is a full-step RK4 advection on (v, b) with u reconstructed from v by continuity at each RK4 substage.
- **Semi-implicit IMEX.** Per time step: (i) extrapolate the nonlinear right-hand side via Adams–Bashforth-2 $f_{\rm nl}^{n+1} \approx 1.5 f^n - 0.5 f^{n-1}$; (ii) solve the $2N_{\rm int} \times 2N_{\rm int}$ linear system $(I - \frac{\Delta t}{2}\mathrm{A})\, U^{n+1} = (I + \frac{\Delta t}{2}\mathrm{A})\, U^n + \Delta t\, [0; f_{\rm nl}^{\rm AB2}]$ per wavenumber, the left-hand matrix pre-factored at setup; (iii) update b by trapezoidal rule linearly and AB2 in advection.
- **Exponential-propagator.** Diagonalise $\mathrm{M} = Q \Lambda Q^{-1}$ per wavenumber at setup. Per step: (i) exact linear half-step to (V, W, B) via analytic integrals $\int_0^{\Delta t/2} V(\tau)\, d\tau$, extending to B by $B \leftarrow B - N^2 \int V$; (ii) RK2 midpoint for the nonlinear advection on (v, b); (iii) second exact linear half-step.

11.5. A.5 IMEX stability derivation

For the scalar test equation $\dot{x} = \mathrm{i}\omega x + \lambda_N x$ with $\lambda_L = \mathrm{i}\omega$ treated by CN and λ_N by AB2, the IMEX(CN, AB2) recurrence is $[x_{n+1}; x_n] = G(z_L, z_N)[x_n; x_{n-1}]$ with G given by (7.1). The spectral radius $\rho(G)$ is the amplification factor; stability requires $\rho(G) \le 1$.

A grid scan over $\omega \Delta t \in [0.01, 3]$ and effective nonlinear coupling $\alpha \in [0, 1.5]$ with three coupling phases (pure imaginary, pure real, and mixed 45°) finds:

- *Imaginary* λ_N: $\rho(G) \approx 1$ for all α; neutrally stable.
- *Real* λ_N: $\rho(G) > 1$ for any $\alpha > 0$, per-step growth $\sim \alpha\omega\Delta t$; unconditionally unstable.
- *Mixed*: intermediate.

At the paper's settings ($\omega = 1.64$, $\Delta t = 0.02$, amp $= 10^{-1}$), the effective α of the $(u \cdot \nabla)v$ coupling yields $\rho(G) \approx 1.005$, giving 800-step amplification $\sim$ 50; the quadratic feedback inherent to the nonlinear block amplifies this to the 10^{79} energy growth observed in Table 7.1.

11.6. A.6 $c(\rho_0, n_g, k_x)$ table

The Proposition 1 constant $c(\rho_0)$ of Section 5.4 is a function of both radial mode index n_g and horizontal wavenumber k_x, through the V_n-weighted measure $d\mu$ in (5.1b). Table A.2 gives the direct measurements on Lane–Emden $n = 3/2$, $\rho_{\rm cut} = 0.05$, $N_y = 96$. In every entry the measured shape gap and the variance prediction (5.1b) agree to all displayed digits.

$k_x/(2\pi/L_y)$	$n_g = 1$	$n_g = 2$	$n_g = 3$	$n_g = 5$	$n_g = 10$
1	19.65	58.27	111.91	277.23	1054.2
2	47.37	117.80	192.63	385.43	1192.4
4	121.24	269.98	402.27	683.45	1632.2
8	317.08	654.64	933.12	1454.3	2843.8

Table A.2: $c(\rho_0, n_g, k_x)$ across 24 (n_g, k_x) modes. The pattern is growth in both n_g (higher radial order, more oscillatory V_n, larger $\mathrm{Var}_\mu(k_x^2 N^2)$) and k_x (scaling closer to k_x^1 than k_x^2, because ω_n^2 also grows with k_x). Every entry is independent of N_y to four significant figures.

12. Appendix B: Code and data availability

All results, figures, and tables reported in this paper are reproducible from the open-source repository

https://github.com/MahoMaho-Rize/stellar2d which contains the reference implementation, the GPU port, and the GYRE benchmark data for the Lane–Emden $n = 3$ polytrope used in Section 4. Each figure and table in the paper is mapped to the script that produces it in the repository's top-level documentation. Runtime environment requirements and dependency versions are documented alongside the code.